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Network meta-analysis for comparative effectiveness of treatments for chronic low back pain disorders: Systematic review protocol

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Network meta-analysis for comparative effectiveness of treatments for chronic low back pain disorders: systematic review protocol

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ABSTRACT

Introduction: Chronic low back pain disorders (CLBDs) present a substantial societal burden; however, optimal treatment remains debated. To date, pairwise and network meta-analyses have evaluated individual treatment modes, yet a comparison of a wide range of common treatments is required to evaluate their relative effectiveness. Using network meta-analysis, we aim to evaluate the effectiveness of treatments (acupuncture, education or advice, electrophysical agents, exercise, manual therapies/manipulation, massage, the McKenzie method, pharmacotherapy, psychological therapies, surgery, epidural injections, percutaneous treatments, traction, physical therapy, multidisciplinary pain management, placebo, "usual care" and/or no treatment) on pain intensity, disability and/or mental health in patients with CLBDs.

Methods and analysis: Six electronic databases and reference lists of 285 prior systematic reviews were searched. Eligible studies will be randomised controlled/clinical trials (including cross-over and cluster designs) that examine individual treatments or treatment combinations in adult patients with CLBDs. Studies must be published in English, German or Chinese as a full-journal publication in a peer-reviewed journal. A narrative approach will be used to synthesise and report qualitative and quantitative data, and, where feasible, network meta-analyses will be performed. Reporting of the review will be informed by Preferred Reporting Items for Systematic Review and Meta-Analysis guidance (PRISMA), including the network meta-analysis extension (PRISMA-NMA). The Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach for network meta-analysis will be implemented for assessing the quality of the findings.

Ethics and dissemination: Ethical approval is not required for this systematic review of the published data. Findings will be disseminated via peer-reviewed publication.

Registration: PROSPERO registration number CRD42020182039.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This study will enable comparison of a wide variety of treatments for chronic low back disorders via network meta-analysis.
- Our study will provide evidence that can be applied in clinical practice and in low back pain management guidelines.
- The quality of evidence will be assessed via the Grading of Recommendations Assessment, Development and Evaluation (GRADE).
- We will address the potential limitation of heterogeneous pathologies being combined into one population by performing sub-group analyses.
- Baseline pain and disability are known to be predictive of outcome and we will account for this in the analysis.

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INTRODUCTION

Low back pain is the greatest cause of disability and lost productivity world-wide [1]. In developed regions, such as the United States of America, Japan, Europe and Australia, the disease generates substantial financial costs [2]. For example, healthcare expenditure is in excess of A$5 billion per year in Australia [3] and US$100 billion per year in the United States of America [3]. The majority of acute cases of back pain resolve without specific intervention, [4] yet chronic low back pain disorders (CLBDs; i.e. >12 weeks duration) generate the greatest proportion of economic burden [5] and affect 20.1 ± 9.8 % of the population worldwide [6]. To reduce the global burden of disease of CLBDs, identifying and implementing the most effective treatment is an urgent priority [7].

To date, pairwise meta-analyses have typically been used to evaluate individual treatment modes for CLBDs [8]. Current recommendations include education, exercise, manual therapy, psychotherapy and multidisciplinary interventions [8,9]. A comparison of a wide range of common treatments and their relative effectiveness for CLBDs is yet to be performed. This evidence would inform management guidelines and clinical decision making. These data would also increase the likelihood that patients receive the most efficacious treatment and/or avoid therapies with similar effectiveness but greater harms. Collectively, this would reduce financial burden at the societal level, as well as improve patient outcome at the individual level.

Network meta-analysis (NMA) permits the ranking of a series of interventions as comparably more or less effective [10,11]. NMA can incorporate data on multiple treatments simultaneously from randomised controlled trials (RCTs) that do not have similar comparator groups by synthesising direct and indirect evidence from a ‘network’ of studies [11–13]. This

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overcomes a key limitation for pairwise meta-analysis, and allows RCTs that do not have a non- or minimal-treatment control group to be included in the analysis [14]. NMA has been used to examine the relative effectiveness of exercise training modalities in non-specific chronic low back pain [15], exercise and education for back pain prevention [16], treatments for lumbar disc herniation [17] and medication for sciatica [18]. However, this approach has not been considered simultaneously for a wide range of common treatments of CLBDs. In this study, we will examine CLBDs, encompassing radicular syndromes and non-specific low back pain [19]. Our primary aim is to determine the relative effectiveness of a variety of common treatments for CLBDs via NMA.

METHODS

This systematic review will be conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [20] and the PRISMA extension for network meta-analyses (PRISMA-NMA) [21]. Covidence (https://www.covidence.org) will be used for article screening and data extraction. This systematic review was prospectively registered on PROSPERO (submitted 24th April 2020; registration number CRD42020182039) prior to initiating data extraction. We will use the PRISMA-P checklist when writing our report [22].

Eligibility criteria

For inclusion, studies will be required to be full peer-reviewed publications (i.e., grey literature including theses and conference abstracts will be excluded) in English, German or Chinese. A meta-epidemiological study by Nussbaumer-Streit et al. [23] found that when non-English

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studies were excluded from systematic reviews of clinical interventions, this had little impact on study conclusions. Furthermore, Cochrane guidelines [24] are ambivalent on the inclusion of non-English language articles and the potential for introduction of bias in reviews. Prior work has suggested that inclusion or exclusion of non-English articles does not influence the effect estimates, but may narrow confidence intervals [25]. We pragmatically chose to include articles in languages in which the author team were fluent. All other inclusion criteria followed the Participants, Interventions, Comparators, Outcomes and Study design (PICOS) framework [21].

Population: Adults (≥ 18 years) with CLBDs. Chronic is defined as pain lasting 12 weeks or more [26]. Since not all studies are consistent in their reporting of pain duration, we will use the following approach: if a study defines it collectively as "chronic", then it will be included. Failing this, if the inclusion criteria of the study are minimum of 12 weeks pain duration or if the median or mean reported duration of pain at baseline in participants is 12 weeks or more, then the study will be included. Recurrent pain (i.e., <12 weeks duration of symptoms and pain-free period of at least 6 months [27]) is excluded. Low back disorder is defined as back pain with or without leg pain where there are no specific spinal pathologies (i.e., vertebral fracture, malignancy, spinal infection, axial spondyloarthritis, cauda equina syndrome [19]). Spondylolisthesis, spondylosis, disc herniation, disc degeneration, scoliosis, deformity (e.g., hemivertebrae) and radicular syndromes (e.g., radicular pain [leg pain or sciatica], radiculopathy, spinal stenosis) are included [19]. “Failed back surgery syndrome” is included as this is not a specific disease [28]. If a study only examines post-surgical pain (e.g., a comparison of management for immediate post-surgical pain as an RCT), we will consider this iatrogenic pain and the study will be excluded.

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Interventions and comparators: The treatment types to be included were determined by the current clinical practice guideline from the American College of Physicians [29] and by the review areas of the Cochrane Back and Neck Group [30]. A detailed list is included in Supplemental Data; however, in brief, we examined acupuncture, education or advice, electrotherapy (including heat and ice electrotherapeutic modalities applied non-invasively), epidural injections, exercise training, manual therapies/manipulation, massage, the McKenzie method, pharmacotherapy, psychological therapies, percutaneous procedures, surgery, traction, physical therapy (otherwise not falling into specific treatment combination), placebo, multidisciplinary pain management, usual care (e.g., general practitioner management), no treatment (true control). Treatment combinations will be considered pending data availability and defined according to their component parts (see Supplemental Data for details) for primary and secondary treatment components. Pending articles included in the review, further sub-group classifications will be considered.

Outcomes: Pain intensity (e.g., VAS, NRS, McGill Pain Questionnaire, or Box scale, other quantitative pain measures), disability (e.g., ODI, RMDQ), mental health (SF-36 MH subscale, depression, anxiety). Adverse events, participant drop-outs and funding sources will be extracted from the included articles.

Study design: Randomised controlled trials, randomised clinical trials, randomised controlled cluster trials, or randomised cross-over trials will be included.

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Search strategy

Six databases (MEDLINE, SPORTDiscus, CINAHL, PsycINFO, EMBASE, CENTRAL) were searched with no restriction on publication dates. The search was initially performed from inception to 14.11.2019 and then was updated on 24.07.2020. Search terms were to find articles on (1) low back disorders and (2) randomised controlled trials (Supplemental Table A). Low back disorder terms included those recommended by the Cochrane Back and Neck review group [31] for non-specific back pain and radicular syndromes [19]. The search terms for identifying RCTs were modelled on Cochrane sensitivity-maximising and precision-maximising search terms to be consistent across databases. Prior systematic reviews in English of any kind of treatment for chronic low back disorders in the last 10 years were screened via a search (January 1990 to July 2019) of MEDLINE, SPORTDiscus, CINAHL, PsycINFO, EMBASE and CENTRAL. Collectively, 285 such systematic reviews were identified. The complete reference lists of these reviews were collated and then screened to remove non-RCTs. Subsequently, 1783 additional references were identified, and after uploading to Covidence, 1008 duplicates were removed, leaving 775 new titles/abstracts. Furthermore, the reference lists of 17 relevant Cochrane reviews not published between January 1990 and July 2019 were screened: 269 additional references were added after discarding 394 duplicates. Following removal of duplicates, a total of 19522 articles remained for screening.

Study selection

For each record, two independent assessors will screen the studies against the predetermined inclusion/exclusion criteria. Disagreements that cannot be resolved amongst the assessors will be addressed by an adjudicator. If unsure, the adjudicator will discuss with the broader study

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team. If still unsure, the study authors will be contacted for clarity. The process for determining study inclusion/exclusion is shown in Figure 1.

Data extraction

For each record, two independent assessors will extract the data. Disagreements that cannot be resolved amongst the assessors will be addressed by an adjudicator. Relevant information pertaining to publication metadata (i.e., author, title, year, journal), study design (e.g., two-arm or multi-arm parallel trial), number of participants, participant characteristics (e.g., age and sex), interventions considered, and outcome measures (pain, disability, mental health, adverse events and funding sources) will be extracted by two independent assessors. Extracted outcome data (pain, disability, mental health) will be pre- and post-intervention mean and standard deviation (SD). When available, data will be extracted for the following time-points: immediate (<1d) effect of treatment, short-term (≥1d but <3mo), intermediate-term (≥3 but <12mo), long-term (≥12mo). Primary and any secondary intervention components will be labelled as per the protocol described in Supplemental Data A.

Data presented as medians or alternate measures of spread will be converted to mean and SD using established formulae [32]. When only figures are presented (rather than numerical data within text), data will be extracted using ImageJ (https://imagej.nih.gov/ij/) to measure the length (in pixels) of the axes to calibrate, and then the length in pixels of the data points of interest [33]. When it is not possible to extract the required data, this information will be requested from the authors at a minimum of three times over a four-week period. Prior to commencing data extraction, this method will be piloted on 30 studies chosen at random. All discrepancies will be referred to an adjudicator.

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Due to the volume of potentially included articles, for each study, information on the population (type of low back pain [non-specific or radicular pain], and sub-population [e.g., 'non-specific', 'low back pain not otherwise stated', 'disc degeneration', 'spondylolisthesis', 'spinal stenosis', 'radiculopathy', 'radicular pain']) and intervention/comparator (intervention duration, free text entry of description of interventions, study-arm labels, primary and secondary intervention classifications (if relevant); see Supplemental Data A) will be extracted first. Then, studies that examine different treatment classes (e.g., exercise versus control, psychological therapies versus exercise, or surgery versus percutaneous therapies; see Supplemental Data A) will be included in subsequent extraction and the remaining studies excluded. This approach will be undertaken because our primary research question concerns different classes of treatments; hence, studies that compare the same class of treatment (e.g., exercise versus exercise, or surgery versus surgery) are less informative for this question.

Risk of bias

Two independent assessors will use the Cochrane Collaboration Risk of Bias [34] to examine potential selection bias (random sequence generation and allocation concealment), performance bias (blinding of patients and personnel), detection bias (blinding of outcome assessment), attrition bias (incomplete outcome data), reporting bias (selective outcome reporting) and other biases. Cluster randomised trials will be assessed as recommended by the Cochrane Collaboration [35]. The revised version of the risk of bias tool [36] will not be used as it was, at initiation of the project, not yet recommended by the Cochrane Collaboration. For each source of bias, studies will be classified as having a low, high or unclear (if reporting was

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not sufficient to assess a particular domain) risk. All discrepancies will be referred to an adjudicator.

Two independent assessors will use the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach for network meta-analysis for assessing the quality of the evidence (Supplemental Data C). We will use a range of equivalence of standardised mean difference (SMD), from -0.5 to 0.5, to evaluate imprecision and inconsistency [37]. Publication bias will be assessed via statistical and non-statistical methods [38]. Indirectness will be judged using Schünemann's approach [39]. Risk of bias will be downgraded by one level if >50% of participants were from studies with selection bias and performance bias. This criterion was selected because inadequate randomization and lack of blinding may lead to an exaggeration of the intervention effect estimates [40–42]. For the categories 'imprecision' and 'inconsistency', we will downgrade by one level if there are some concerns and two levels if there are major concerns. Indirectness will be downgraded by one level if deemed serious and two levels if deemed very serious. We will downgrade one level if publication bias is suspected. The GRADE approach [43,44] will also be used to assess the quality of the evidence of pair-wise comparisons. All discrepancies will be referred to an adjudicator.

Statistical analyses

When studies are reverse scaled (i.e., higher values indicated better outcomes rather than lower values), the mean in each group will be multiplied by -1 as recommended in the Cochrane Handbook. As all of the outcomes of interest will be continuous or ordinal, but could be measured on different scales, SMD will be used as the effect estimates [45]. A

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minimum of 50 participants will be required per class of treatment for it to be included in meta-analysis. We have limited the number of participants to try to limit the impacts of small study effects on the results of any particular class [45]. Furthermore, because we are conducting an analysis of Standardised Mean Differences (SMD), small study effects are likely to be exacerbated as both the mean and the standard deviations are likely to be estimated with greater variability in small studies, and for SMD both of these contribute to the treatment effect. To further investigate our choice of SMD as an effect measure, we will conduct sensitivity analyses with internal reference baseline SDs for each scale [46].

Where a study does not report data in a form where the SD can be extracted or calculated [32], and authors are not able to fulfil data requests, SDs will be imputed and their impact evaluated in sensitivity analyses. To impute missing SDs, we will perform a regression of log(SD) on log(mean) in studies reporting SD following the approach of Marinho et al. 2003, adjusting for measurement scale and follow-up time [47]. We will then use this regression model to predict SDs that are missing.

Cluster randomised trials will be included in the analysis as per Cochrane guidance. Sensitivity analysis will be conducted in pairwise analyses with a range of different ICCs to check the robustness of the results. [48] For crossover trial designs, we will include the estimated relative treatment effect from the study where possible, where the authors have tested for carryover effects and found no evidence of this. Where this is not the case, we will only include the first period of the crossover trial. In time-course Model-Based Network Meta Analyses (MBNMA), only the inclusion of the first time-period will be possible.

Network Meta-Analysis

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Bayesian network meta-analysis (NMA) will be performed at discrete time-points (immediate (<1d) effect of treatment, short-term (≥1d but <3mo), intermediate-term (≥3 but <12mo), and long-term (≥12mo)) using the R (r-project.org) package multinma [49]. Time-course MBNMA will be conducted using the R package MBNMAtime [50,51]. This package enables the incorporation of multiple time-points per study in Bayesian NMA to inform estimates of effect size over time. Network connectivity will be explored via network plots. Network plots help to visualise how the evidence in the network is connected and allow identification of which studies compare which treatments. This aids in understanding which treatment effects can be estimated. The time-course relationship will be examined by a time plot, which is a plot of the raw study responses over time. Time plots help to elucidate the underlying time course of the treatment effects and help to identify which statistical time model is appropriate.

Where data allow and where there is a plausible clinical reason for doing so, treatment effects will be assumed to be common or exchangeable within a class. This allows for treatments to be nested within a class, which relaxes assumptions regarding the similarity of interventions whilst improving network connectivity [13]. We will use the deviance information criterion (DIC) to compare the different models (common/exchangeable class effect models, time-course models) to assess their parsimony [52].

For standard NMA models we will rank the relative effects of each treatment/class, and for time-course MBNMA models we will rank the relative effects of each treatment/class for each time-course parameter. We will also rank the full area under the time-course function for each treatment/class at 0–3 months, 0–6 months and 0–12 months. Cumulative rankograms will be plotted; these show the range of rankings of different treatments/classes for each ranked parameter. Sensitivity of model results to the choice of prior distributions will be investigated.

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Assessing key assumptions of pairwise and network meta-analysis

The authors recommended a strong and rigorous focus on the evaluation of the similarity and homogeneity assumptions.

Assessment of similarity and homogeneity assumptions

A qualitative assessment of the clinical similarity of the different populations and treatments will be performed by important variables such as baseline pain intensity, baseline disability and pain duration. Between-study SD will be estimated and reported from random effects models, and the impacts of subgrouping or meta-regression on this will be examined. Pair-wise meta-analysis of data will be synthesised via SMDs with accompanying 95% confidence intervals using a frequentist random effects model with a restricted maximum likelihood estimator for the between study variance Tau^2. These analyses will be carried out with the R package "metafor" [53]. Visual inspection of the forest plots, statistical estimates of heterogeneity (I^2, Tau) and 95% prediction intervals will be used to assess the validity of homogeneity assumptions. Small study effects and publication bias will be assessed for each pairwise comparison by visual inspection of the contour-enhanced funnel plot. Outlier and influential study analysis will be performed with metafor for pairwise meta-analyses to further detect potential heterogeneity [54]. Meta-regression with potential effect modifiers (pre-intervention pain severity and disability, baseline psychological conditions, presence of cointerventions and type of low back pain); [55–57] will be used to further check for potential heterogeneity among the pairwise comparisons [58].

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In the presence of effect modification in pairwise comparisons (identified using meta-regression), we will also fit network meta-regression with these potential effect modifiers for NMAs conducted at each time-point using the package multinma [49].

Consistency assumptions

For the Bayesian approach, consistency assumptions will be first checked via an unrelated mean effects (UME) model which does not assume consistency [59]. The UME model only synthesises direct relative effects between each arm in a study and the study reference treatment. If the consistency assumption holds then the results from the UME and NMA models will be similar. Changes in between-study SD or residual deviance are also suggestive of inconsistency. If comparison between UME and NMA models is suggestive of inconsistency, node-splitting will be performed [60]. In node-splitting, network contrasts are split into direct and indirect evidence contributions, which can then be compared to examine their similarity.

Additional assumptions required for analysis of time-course data

Given that data will be reported at different follow-up times in different studies, information is unlikely to be available for all treatments at all time-points of interest. For this reason, additional assumptions regarding specific parameters for treatments/classes may be required. For example, in the case of a treatment for which information is only available at shorter follow-up times, explicit assumptions regarding its long-term efficacy will be required. The treatment's long-term efficacy could be assumed to be the same as (or similar to) that of another treatment in the network that might have a similar mechanism of action (e.g., within the same class), for which long-term data is available. Alternatively, it could be assigned a specific value or an informative prior as determined by clinical expertise. In such an example, long-term

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results for this treatment will therefore be sensitive to these assumptions, and results will be interpreted accordingly [51]. Assumptions made in this way will be clearly stated and justified.

Subgroup and sensitivity analyses

Pending data availability, we will perform subgroup analyses to explore whether inconsistency/heterogeneity and group differences in the outcomes are influenced by type of low back disorder (e.g., non-specific chronic low back pain, radicular syndrome), type of treatment (e.g., surgical, pharmacological) or by exclusion of the multidisciplinary node and the physical therapy (otherwise not falling into specific treatment combination) node from analyses. The treatment node may be a source of significant heterogeneity/inconsistency for the overall NMA due to the variability of this treatment definition compared to other interventions. Subgroup analysis focussing on key participant or study characteristics can produce smaller, more homogenous networks and can be a good strategy to analyse inconsistency/heterogeneity with fewer assumptions and pitfalls then NMA meta-regression [61]. If we are unable to identify the source of inconsistency, we will highlight that this limits the usefulness of the analysis for drawing meaningful conclusions in such a heterogeneous population.

Further, pending data-availability, we will consider the following sensitivity analyses

- Excluding studies with imputed missing standard deviations and imputed medians.
- Study sample size: impact of studies including less than 20 participants in all study-arms.
- Drop out numbers and handling of dropouts within studies: the impact of the proportion of dropouts (if reported) and the kind of analysis individual studies performed (e.g.,

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analysing all participants using imputation of missing data vs. analysing complete cases only).

- Comparison of class effect models to a model with fully independent treatment effects that assume no within-class similarity, to assess the statistical validity of class assumptions.
- Secondary treatment components (see Supplemental Data A): the impact of treatment combinations where secondary classes of treatment are present in all arms will be considered by fitting models that incorporate combinations as different nodes in the network. This can be used to assess the assumption of additivity of combined treatments. We will also investigate the impact of ordering of primary/secondary treatment components by fitting a model in which the order is ignored (e.g. "Physical therapy + massage" assumed to be equivalent to "Massage + physical therapy")
- Excluding unclear generic nodes (e.g., physical therapy otherwise not falling into specific treatment combination)
- Risk of bias: To examine the influence of specific studies/comparisons on the treatment rankings we will conduct a threshold analysis where possible [51] using the R package nmathresh.
- Choice of SMD as an effect measure by using internal reference baseline SDs for analysis. [46]

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DISCUSSION

This NMA will determine the relative effectiveness of a variety of common treatments for CLBDs. Conducting NMA on this topic constitutes a shift towards the highest level of medical evidence [62]. Our NMA has a much broader scope than prior work, such as that concerned solely with pharmacotherapy [63–66], exercise training [15,67,68], traditional Chinese medicine [69], or psychotherapy [70]. Moreover, the broad inclusion criteria and number of interventions considered in our NMA will result in a greater number of included interventions than previous broad NMAs that examined non-pharmacotherapy [71] and surgery-based interventions [72], which included 31 and 12 interventions, respectively. The breadth of our NMA is important given that CLBDs are inherently heterogenous, yet progenitors do not influence decision making regarding treatment sought [73]. For this reason, CLBDs (excluding specific causes) are commonly treated in line with generic clinical guidelines [74]. This underpins the importance of our NMA, as these guidelines do not distinguish whether one treatment is superior to another for this collective of patients with chronic pain. Given the lack of evidence that treatment efficacy differs by underlying pain progenitor, we believe it is reasonable to assume exchangeability of these studies and transitivity within the network in terms of population. Other than recent suggestions that machine learning [75] may one day identify evidence-based sub-groups that respond 'better' to specific treatments, we surmise that our NMA will markedly contribute to overcoming current limitations in the management of CLBDs pertaining to treatment decision making.

To our knowledge, there is only one other NMA currently being conducted with a similar scope to our protocol [76]. Our NMA overcomes several cardinal limitations of this protocol: (1) we consider CLBD, rather than solely non-specific low back pain; (2) we consider additional languages for article inclusion, rather than English only; and (3) our treatment classification is

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more nuanced, rather than simplistic (e.g., the other protocol typically considers two types of treatment within a particular class). Of note, we registered our systematic review prior to publication of this other protocol, and it is unclear when their work is due to be published.

Despite the many strengths of our proposed NMA, we would be remiss not to acknowledge potential limitations. First, due to the inclusion of radicular syndromes in the patient population, it might be necessary to analyse this population in different networks/subsets because the presence of this may be an effect modifier [77] and lead to intransitivity. Second, we do not consider multicomponent interventions in our statistical model, which might have an impact on the estimates [78,79]. By ignoring additional treatment components given in both arms of included studies, we assume additivity of different treatment components. While we will investigate the effects of this (see *Sensitivity Analyses*), fully accounting for it by modelling all combinations of treatments as separate interventions is likely to lead to disconnected networks of evidence, which poses its own problem for evidence synthesis and decision making [80]. Third, while we propose a variety of subgroup analyses to investigate the impact of effect modification, potential effect modifiers may be poorly reported in many studies. However, there is no clear evidence of important effect modification in CLBD to date. As pointed out in the recent Lancet Low Back Pain Series [8], relative treatment efficacy for different kinds of interventions appears (to date) to be surprisingly similar. Fourth, usual care may vary between included studies (e.g., authors' stance on whether or not usual analgesic pharmacotherapy was permitted), yet given few studies in the CLBD field employ methods of strict observation, we surmise that the majority, if not all, of existing studies are inherently at risk of this form of bias. Finally, as with all meta-analyses, dealing with co-interventions has implicit complexities. Our decision to consider interventions that combine multiple forms of interventions of interest may impede our capacity to differentiate the effects of one individual treatment. However, we

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contend that this approach allows for the inclusion of more trials that, when compared to a strict approach that excluded any interventions with co-intervention, reflects more realistically the realities of clinical practice. This, in our view, leads to less potential bias (e.g. inclusion of studies that simply failed to report co-interventions) and greater confidence in our effect estimates.

In conclusion, the current project will enable a significant advance in synthesising knowledge on the comparative effectiveness of a wide variety of treatments for chronic low back disorders. This has, to date, not been performed and will inform patient management and clinical practice guidelines.

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References

1 Vos T, Flaxman AD, Naghavi M, *et al.* Years lived with disability (YLDs) for 1160 sequelae of 289 diseases and injuries 1990-2010: a systematic analysis for the Global Burden of Disease Study 2010. *Lancet Lond Engl* 2012;**380**:2163–96. doi:10.1016/S0140-6736(12)61729-2

2 Dagenais S, Caro J, Haldeman S. A systematic review of low back pain cost of illness studies in the United States and internationally. *Spine J* 2008;**8**:8–20. doi:10.1016/j.spinee.2007.10.005

3 Martin BI, Deyo RA, Mirza SK, *et al.* Expenditures and health status among adults with back and neck problems. *JAMA* 2008;**299**:656–64. doi:10.1001/jama.299.6.656

4 da C Menezes Costa L, Maher CG, Hancock MJ, *et al.* The prognosis of acute and persistent low-back pain: a meta-analysis. *CMAJ* 2012;**184**:E613-624. doi:10.1503/cmaj.111271

5 Maetzel A, Li L. The economic burden of low back pain: a review of studies published between 1996 and 2001. *Best Pract Res Clin Rheumatol* 2002;**16**:23–30. doi:10.1053/berh.2001.0204

6 Hoy D, Bain C, Williams G, *et al.* A systematic review of the global prevalence of low back pain. *Arthritis Rheum* 2012;**64**:2028–37. doi:10/fzbtrz

7 Hartvigsen J, Hancock MJ, Kongsted A, *et al.* What low back pain is and why we need to pay attention. *The Lancet* 2018;**391**:2356–67. doi:10/gdnd74

8 Maher C, Underwood M, Buchbinder R. Non-specific low back pain. *The Lancet* 2017;**389**:736–47. doi:10/gcpzvb

9 Almeida M, Saragiotto B, Richards B, *et al.* Primary care management of non-specific low back pain: key messages from recent clinical guidelines. *Med J Aust* 2018;**208**:272–5. doi:10.5694/mja17.01152

10 Shim S, Yoon B-H, Shin I-S, *et al.* Network meta-analysis: application and practice using Stata. *Epidemiol Health* 2017;**39**:e2017047. doi:10.4178/epih.e2017047

11 Mills EJ, Thorlund K, Ioannidis JPA. Demystifying trial networks and network meta-analysis. *BMJ* 2013;**346**:f2914. doi:10.1136/bmj.f2914

12 Dias S, Welton NJ, Sutton AJ, *et al. NICE DSU Technical Support Document 2: A Generalised Linear Modelling Framework for Pairwise and Network Meta-Analysis of Randomised Controlled Trials*. London: : National Institute for Health and Care Excellence (NICE) 2014. http://www.ncbi.nlm.nih.gov/books/NBK310366/ (accessed 8 Dec 2020).

13 Dias S, editor. *Network meta-analysis for decision making*. Hoboken, NJ: : Wiley 2018.

14 Tagliaferri SD, Miller CT, Ford JJ, *et al.* Randomized Trial of General Strength and Conditioning Versus Motor Control and Manual Therapy for Chronic Low Back Pain on Physical and Self-Report Outcomes. *J Clin Med* 2020;**9**. doi:10.3390/jcm9061726

15 Owen PJ, Miller CT, Mundell NL, *et al.* Which specific modes of exercise training are most effective for treating low back pain? Network meta-analysis. *Br J Sports Med* 2019;:in press. doi:10.1136/bjsports-2019-100886

16 Huang R, Ning J, Chuter VH, *et al.* Exercise alone and exercise combined with education both prevent episodes of low back pain and related absenteeism: systematic review and network meta-analysis of randomised controlled trials (RCTs) aimed at preventing back pain. *Br J Sports Med* Published Online First: 31 October 2019. doi:10.1136/bjsports-2018-100035

17 Arts MP, Kuršumović A, Miller LE, *et al.* Comparison of treatments for lumbar disc herniation: Systematic review with network meta-analysis. *Medicine (Baltimore)* 2019;**98**:e14410. doi:10.1097/MD.0000000000014410

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

18 Guo J-R, Jin X-J, Shen H-C, *et al.* A Comparison of the Efficacy and Tolerability of the Treatments for Sciatica: A Network Meta-Analysis. *Ann Pharmacother* 2017;**51**:1041–52. doi:10.1177/1060028017722008
19 Bardin LD, King P, Maher CG. Diagnostic triage for low back pain: a practical approach for primary care. *Med J Aust* 2017;**206**:268–73. doi:10.5694/mja16.00828
20 Moher D, Liberati A, Tetzlaff J, *et al.* Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;**6**:e1000097. doi:10.1371/journal.pmed.1000097
21 Hutton B, Salanti G, Caldwell DM, *et al.* The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Ann Intern Med* 2015;**162**:777–84. doi:10.7326/M14-2385
22 Moher D, Shamseer L, Clarke M, *et al.* Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. *2015*;**4**:1. doi:10/gcsk56
23 Nussbaumer-Streit B, Klerings I, Dobrescu AI, *et al.* Excluding non-English publications from evidence-syntheses did not change conclusions: a meta-epidemiological study. *J Clin Epidemiol* 2020;**118**:42–54. doi:10/ghpwpp
24 Higgins J, Thomas J, Chandler J, *et al.*, editors. *Cochrane handbook for systematic reviews of interventions version 6.2 (updated February 2021)*. Cochrane 2021. https://training.cochrane.org/handbook (accessed 16 Aug 2021).
25 Morrison A, Polisena J, Husereau D, *et al.* The effect of English-language restriction on systematic review-based meta-analyses: a systematic review of empirical studies. *Int J Technol Assess Health Care* 2012;**28**:138–44. doi:10.1017/S0266462312000086
26 Koes BW, van Tulder M, Lin C-WC, *et al.* An updated overview of clinical guidelines for the management of non-specific low back pain in primary care. *Eur Spine J* 2010;**19**:2075–94. doi:10.1007/s00586-010-1502-y
27 van Tulder M, Becker A, Bekkering T, *et al.* Chapter 3. European guidelines for the management of acute nonspecific low back pain in primary care. *Eur Spine J* 2006;**15 Suppl 2**:S169-191. doi:10.1007/s00586-006-1071-2
28 Clancy C, Quinn A, Wilson F. The aetiologies of Failed Back Surgery Syndrome: A systematic review. *J Back Musculoskelet Rehabil* 2017;**30**:395–402. doi:10.3233/BMR-150318
29 Qaseem A, Wilt TJ, McLean RM, *et al.* Noninvasive Treatments for Acute, Subacute, and Chronic Low Back Pain: A Clinical Practice Guideline From the American College of Physicians. *Ann Intern Med* 2017;**166**:514–30. doi:10.7326/M16-2367
30 Cochrane Back and Neck. Our reviews. 2020.https://back.cochrane.org/our-reviews (accessed 17 Jun 2020).
31 Furlan AD, Malmivaara A, Chou R, *et al.* 2015 Updated Method Guideline for Systematic Reviews in the Cochrane Back and Neck Group. *Spine* 2015;**40**:1660–73. doi:10.1097/BRS.0000000000001061
32 Wan X, Wang W, Liu J, *et al.* Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol* 2014;**14**:135. doi:10/f7xmgr
33 Vucic K, Jelicic Kadic A, Puljak L. Survey of Cochrane protocols found methods for data extraction from figures not mentioned or unclear. *J Clin Epidemiol* 2015;**68**:1161–4. doi:10.1016/j.jclinepi.2014.11.016
34 Higgins JPT, Altman DG, Gotzsche PC, *et al.* The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;**343**:d5928–d5928. doi:10/fcq7fv
35 Higgins J, Deeks JJ, Altman DG. Chapter 16: Assessing risk of bias in cluster-randomized trials. In: Higgins J, Green, S, eds. *Cochrane Handbook for Systematic Reviews*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of Interventions. The Cochrane Collaboration 2011. https://handbook-5-1.cochrane.org/chapter_16/16_3_2_assessing_risk_of_bias_in_cluster_randomized_trials.htm (accessed 16 Jul 2021).

36 Sterne JAC, Savović J, Page MJ, *et al.* RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;:l4898. doi:10/ggmfc5

37 Norman GR, Sloan JA, Wyrwich KW. Interpretation of changes in health-related quality of life: the remarkable universality of half a standard deviation. *Med Care* 2003;**41**:582–92. doi:10/dhstrr

38 Nikolakopoulou A, Higgins JPT, Papakonstantinou T, *et al.* CINeMA: An approach for assessing confidence in the results of a network meta-analysis. *PLOS Med* 2020;**17**:e1003082. doi:10/ggthw8

39 Schünemann H, Higgins J, GE Vist, *et al.* Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins J, Thomas J, Chandler J, *et al.*, eds. *Cochrane Handbook for Systematic Reviews of Interventions*. The Cochrane Collaboration 2021. https://training.cochrane.org/handbook/current/chapter-14 (accessed 16 Aug 2021).

40 Savović J, Turner RM, Mawdsley D, *et al.* Association Between Risk-of-Bias Assessments and Results of Randomized Trials in Cochrane Reviews: The ROBES Meta-Epidemiologic Study. *Am J Epidemiol* 2018;**187**:1113–22. doi:10/gdgm64

41 Armijo-Olivo S, Fuentes J, da Costa BR, *et al.* Blinding in Physical Therapy Trials and Its Association with Treatment Effects: A Meta-epidemiological Study. *Am J Phys Med Rehabil* 2017;**96**:34–44. doi:10/gmhbgv

42 Armijo-Olivo S, Saltaji H, Costa BR da, *et al.* What is the influence of randomisation sequence generation and allocation concealment on treatment effects of physical therapy trials? A meta-epidemiological study. *BMJ Open* 2015;**5**:e008562. doi:10/gb5db8

43 Puhan MA, Schünemann HJ, Murad MH, *et al.* A GRADE Working Group approach for rating the quality of treatment effect estimates from network meta-analysis. *BMJ* 2014;**349**:g5630. doi:10/gf9bh2

44 Brignardello-Petersen R, Bonner A, Alexander PE, *et al.* Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis. *J Clin Epidemiol* 2018;**93**:36–44. doi:10/gcz4pg

45 Sterne JA, Gavaghan D, Egger M. Publication and related bias in meta-analysis: power of statistical tests and prevalence in the literature. *J Clin Epidemiol* 2000;**53**:1119–29. doi:10/fjt66s

46 Daly C, Welton NJ, Dias S, *et al.* Meta-Analysis of Continuous Outcomes. NICE Guidelines Technical Support Unit http://www.bristol.ac.uk/media-library/sites/social-community-medicine/documents/mpes/gmd-2-continuous-jan2021.pdf

47 Marinho VC, Higgins JP, Sheiham A, *et al.* Fluoride toothpastes for preventing dental caries in children and adolescents. *Cochrane Database Syst Rev* 2003;:CD002278. doi:10/c9rtvd

48 Higgins JP, Eldridge S, Li T. Chapter 23: Including variants on randomized trials. In: Higgins J, Thomas J, Chandler J, *et al.*, eds. *Cochrane Handbook for Systematic Reviews of Interventions*. Cochrane 2021. https://training.cochrane.org/handbook/current/chapter-23 (accessed 16 Jul 2021).

49 Phillippo D. *dmphillippo/multinma: v0.3.0*. Zenodo 2021. doi:10.5281/zenodo.4617744

50 Pedder H. *MBNMAtime: Run time-course MBNMA models*. https://CRAN.R-project.org/package=MBNMAtime

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

51 Pedder H, Dias S, Bennetts M, *et al.* Modelling time-course relationships with multiple treatments: Model-based network meta-analysis for continuous summary outcomes. *Res Synth Methods* 2019;**10**:267–86. doi:10/ghnwjc

52 Spiegelhalter DJ, Best NG, Carlin BP, *et al.* Bayesian measures of model complexity and fit. *J R Stat Soc Ser B Stat Methodol* 2002;**64**:583–639. doi:10/dfrgt6

53 Viechtbauer W. Conducting Meta-Analyses in R with the metafor Package. *J Stat Softw* 2010;**36**. doi:10/gckfpj

54 Viechtbauer W, Cheung MW-L. Outlier and influence diagnostics for meta-analysis. *Res Synth Methods* 2010;**1**:112–25. doi:10/fs53xp

55 Boonstra AM, Reneman MF, Waaksma BR, *et al.* Predictors of multidisciplinary treatment outcome in patients with chronic musculoskeletal pain. *Disabil Rehabil* 2015;**37**:1242–50. doi:10.3109/09638288.2014.961657

56 Cecchi F, Pasquini G, Paperini A, *et al.* Predictors of response to exercise therapy for chronic low back pain: result of a prospective study with one year follow-up. *Eur J Phys Rehabil Med* 2014;**50**:143–51.

57 van der Hulst M, Vollenbroek-Hutten MMR, Ijzerman MJ. A systematic review of sociodemographic, physical, and psychological predictors of multidisciplinary rehabilitation-or, back school treatment outcome in patients with chronic low back pain. *Spine* 2005;**30**:813–25. doi:10.1097/01.brs.0000157414.47713.78

58 Donegan S, Williamson P, D'Alessandro U, *et al.* Assessing key assumptions of network meta-analysis: a review of methods: Assessing assumptions of network meta-analysis. *Res Synth Methods* 2013;**4**:291–323. doi:10/gf8ddc

59 Dias S, Welton NJ, Sutton AJ, *et al. NICE DSU Technical Support Document 4: Inconsistency in Networks of Evidence Based on Randomised Controlled Trials*. London: : National Institute for Health and Care Excellence (NICE) 2014. http://www.ncbi.nlm.nih.gov/books/NBK310372/ (accessed 8 Dec 2020).

60 van Valkenhoef G, Dias S, Ades AE, *et al.* Automated generation of node-splitting models for assessment of inconsistency in network meta-analysis. *Res Synth Methods* 2016;**7**:80–93. doi:10/f8t6mt

61 Wade R, Sharif-Hurst S, Dias S. Patient characteristics as effect modifiers for psoriasis biologic treatment response: an assessment using network meta-analysis subgroups. *Syst Rev* 2020;**9**:132. doi:10/gmc79b

62 Faltinsen EG, Storebø OJ, Jakobsen JC, *et al.* Network meta-analysis: the highest level of medical evidence? *BMJ Evid-Based Med* 2018;**23**:56–9. doi:10/ghn2kn

63 Bagg MK, McLachlan AJ, Maher CG, *et al.* Paracetamol, NSAIDS and opioid analgesics for chronic low back pain: a network meta-analysis. *Cochrane Database Syst Rev* 2018;**2018**:CD013045. doi:10/gjkmg7

64 Migliorini F, Maffulli N, Eschweiler J, *et al.* Non-steroidal anti-inflammatory drugs and gabapentinoids for chronic lumbar pain: a Bayesian network meta-analysis of randomized controlled trials. *Br Med Bull* 2021;**138**:85–95. doi:10.1093/bmb/ldab003

65 Migliorini F, Maffulli N, Baroncini A, *et al.* Opioids for chronic low back pain management: a Bayesian network meta-analysis. *Expert Rev Clin Pharmacol* 2021;**14**:635–41. doi:10.1080/17512433.2021.1903316

66 Boya C, Bansal D, Kanakagiri S, *et al.* Efficacy and Safety of Opioid Analgesics for the Management of Chronic Low Back Pain: An Evidence from Bayesian Network Meta-Analysis. *Pain Physician* 2021;**24**:73–82.

67 Anheyer D, Klose P, Koch AK, *et al.* Comparative efficacy of different exercise interventions in chronic non-specific low back pain: protocol of a systematic review and network meta-analysis. *BMJ Open* 2020;**10**:e036050. doi:10.1136/bmjopen-2019-036050

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

68 Gao C, Chen G, Yang H, *et al.* Relative effectiveness of different forms of exercises for treatment of chronic low back pain: protocol for a systematic review incorporating Bayesian network meta-analysis. *BMJ Open* 2019;**9**:e025971. doi:10.1136/bmjopen-2018-025971

69 Yu H, Wang H, Ma T, *et al.* TCM nonpharmacological interventions for chronic low-back pain: A protocol for systematic review and network meta-analysis. *Medicine (Baltimore)* 2020;**99**:e22547. doi:10/ghmvnf

70 Ikemoto T, Miki K, Matsubara T, *et al.* Psychological Treatment Strategy for Chronic Low Back Pain. *Spine Surg Relat Res* 2018;**3**:199–206. doi:10/gmhbgw

71 Sun W, Zhang H, Lv C, *et al.* Comparative efficacy of 12 non-drug interventions on non-specific chronic low back pain in nurses: A systematic review and network meta-analysis. *J Back Musculoskelet Rehabil* 2021;**34**:499–510. doi:10/gmc79c

72 Rihn JA, Radcliff K, Norvell DC, *et al.* Comparative Effectiveness of Treatments for Chronic Low Back Pain: A Multiple Treatment Comparison Analysis. *Clin Spine Surg* 2017;**30**:204–25. doi:10/gjkk7d

73 Maher C, Underwood M, Buchbinder R. Non-specific low back pain. *Lancet Lond Engl* 2017;**389**:736–47. doi:10.1016/S0140-6736(16)30970-9

74 Oliveira CB, Maher CG, Pinto RZ, *et al.* Clinical practice guidelines for the management of non-specific low back pain in primary care: an updated overview. *Eur Spine J* 2018;**27**:2791–803. doi:10/gfp76w

75 Tagliaferri SD, Angelova M, Zhao X, *et al.* Artificial intelligence to improve back pain outcomes and lessons learnt from clinical classification approaches: three systematic reviews. *Npj Digit Med* 2020;**3**:1–16. doi:10/ghn2mv

76 Thompson T, Dias S, Poulter D, *et al.* Efficacy and acceptability of pharmacological and non-pharmacological interventions for non-specific chronic low back pain: a protocol for a systematic review and network meta-analysis. *Syst Rev* 2020;**9**:130. doi:10/ghmvnm

77 Low back pain and sciatica in over 16s: assessment and management. https://www.nice.org.uk/guidance/ng59 (accessed 31 Jul 2021).

78 Welton NJ, Caldwell DM, Adamopoulos E, *et al.* Mixed treatment comparison meta-analysis of complex interventions: psychological interventions in coronary heart disease. *Am J Epidemiol* 2009;**169**:1158–65. doi:10/c29ps8

79 Rücker G, Petropoulou M, Schwarzer G. Network meta-analysis of multicomponent interventions. *Biom J Biom Z* 2020;**62**:808–21. doi:10/ggd9ff

80 Stevens JW, Fletcher C, Downey G, *et al.* A review of methods for comparing treatments evaluated in studies that form disconnected networks of evidence. *Res Synth Methods* 2018;**9**:148–62. doi:10/gcpj92

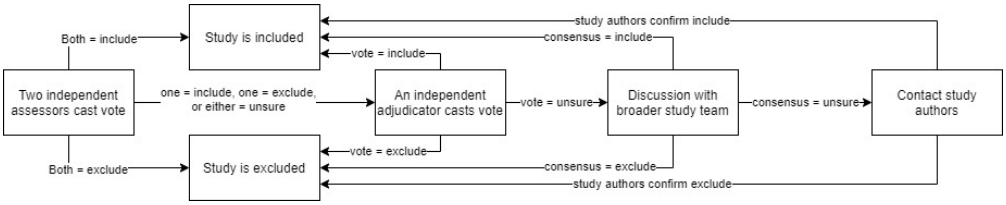
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Figure 1. The process for determining study inclusion/exclusion.

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Process for determining study inclusion/exclusion.

324x63mm (72 x 72 DPI)

Supplemental Data A: Definitions of interventions and primary/secondary interventions

Acupuncture (acu)

Per prior Cochrane review [1], the definition of acupuncture used was "*the diagnosis was made using traditional acupuncture theory and the needles were inserted in classical meridian points, extra points or ah-shi points (painful points)*". Dry needling was classified with acupuncture and required needles to be inserted into myofascial trigger points. Acupressure, laser acupuncture and acupuncture via electrical stimulation were excluded from this acupuncture group as they did not involve needling. These interventions were included, respectively, under massage (acupressure) and electrotherapies (laser and electrical acupuncture).

Treatments within Class:

- acu_need: acupuncture following (traditional) acupuncture theory
- acu_dry: dry needling

Education (edu)

Patient education has been defined [2] previously "*a systematic experience, in a one-to one situation, that consists of one or more methods, such as the provision of information and advice and behaviour modification techniques*". Similar to this prior review, we considered education to occur when back pain patients were given information to help them understand their condition, what behaviours are likely to be more beneficial. 'Back school' interventions were considered education. Advice to stay active was considered education. Both group and individual education were included. Using brochure or booklet with education material was included if a clinician explained the information to the patient. Studies on instructions as to how to perform other kinds of interventions (e.g., how to do exercise, or were included, studies on instructions on how to perform exercises were not included.

Treatments within Class:

- edu_school: back school
- edu_pne: pain neuroscience education
- edu_book: via printed materials
- edu_grpind: remaining group and individual education

Electrophysical agents (elc)

Therapeutic heat and cold, laser (including laser acupuncture) and light therapies, classic electrotherapies (e.g., electrical stimulation modalities including TENS; electrical acupuncture also included here), various electromagnetic applications (e.g., pulsed shortwave therapy), ultrasound therapy and a variety of mechanical therapies (e.g., vibration therapy and intermittent pneumatic compression therapy) are included as electrophysical agents given these modalities are considered comparable [3]. The electrophysical agents must be applied externally without breaking or piercing the skin.

Whole body vibration, where a person experiences vibration through their whole body, is excluded.

Treatments within Class:

- elc_electric: electrical stim or input of some form
- elc_hot: heat
- elc_cold: cold

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- elc_mech: ultrasound therapy and a variety of mechanical therapies
- elc_etc: magnetic and remaining included

Epidural injections (epi)

As per prior Cochrane review [4], epidural injections involve the delivery of corticosteroid medication to the epidural space via injection. The anatomical approaches considered included, but were not limited to: caudal, interlaminar, and transforaminal approaches.

Treatments within Class:

- epi_caud: caudal approach
- epi_inter: interlaminar approach
- epi_trans: transforaminal approach
- epi_other: other included epidural INT not included in anatomical approaches listed above

Exercise (exe)

Exercise therapy has been [5] defined as "*a series of specific movements with the aim of training or developing the body by a routine practice or as physical training to promote good physical health*". We required that a clinician or study investigator instructed and/or prescribed exercises to patients with the goal of improving the patient's back disorder. Exercise could be performed as a group or individually. Whole body vibration and whole body vibration exercise was excluded.

Treatments within Class:

- exe_res: resistance exericse
- exe_sta: stabilization_motor_control
- exe_eso: pilates, yoga, traditional eastern approaches
- exe_aer: aerobic (e.g cycling, walking)
- exe_str: streching
- exe_oth: other and water based

Manual therapies and manipulation (man)

A prior Cochrane review [6] defined mobilisation as the "*use low-grade velocity, small or large amplitude passive movement techniques within the patient's range of motion and control*" and manipulation as "*a high velocity impulse or thrust applied to a synovial joint over a short amplitude at or near the end of the passive or physiologic range of motion, which is often accompanied by an audible crack*". The term "adjustments" is sometimes used in conjunction with chiropractic or osteopathic manual therapy treatment. Studies that incorporate visceral techniques as part of an osteopathy intervention will be included.

Treatments within Class:

- man_man: manual therapy and mobilisation (without manipulation)
- man_mip: manipulation
- man_chos: chiropractic or osteopathy not otherwise more precisely specified

Massage (mas)

Massage has been [7] defined as "*the manipulation of the soft tissue of whole body areas to bring about generalised improvements in health, such as relaxation or improved sleep, or*

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specific physical benefits, such as relief of muscular aches and pains" Trigger point therapy, myofascial release, Shiatsu, reflexology, and acupressure are also classified as massage.

Treatments within Class:

- mas_mas: massage
- mas_tpm: Trigger point therapy, myofascial release
- mas_oth: Shiatsu, reflexology, acupressure and other specifically named treatments determined to be massage

McKenzie (mck)

The McKenzie method [8] has also been termed Mechanical Diagnosis and Therapy and is a system that involves the use of mechanical loading strategies to guide specific treatment based on the patient's responses to these mechanical loading strategies (sub-group membership) [9]. In this treatment approach, treatment is individualized for each patient based on the response of their pain/impairment to mechanical loading strategies (sustained or repeated movements and postures) and classified into dysfunction, posture and derangement syndromes. Given it is the most prevalent classification, studies using directional preference treatment only (for derangement syndrome) will also be included. Directional preference management was defined as individualized treatment based on the response to mechanical loading strategies. Trials evaluating the effect of directional preference management on back pain were included.

Treatments within Class:

- mck_mck: Mckenzie

Pharmacotherapy (pha)

Pharmacotherapy interventions considered in this review included non-steroidal anti-inflammatory drugs (NSAIDs), Opioids, Skeletal muscle relaxants, Benzodiazepines, Antidepressants, Acetaminophen (paracetamol), systemic corticosteroids and anticonvulsants. Analgesic medicines work in various ways to reduce the intensity of pain but may also cause unwanted harmful effects.

Treatments within Class:

- pha_nsai: NSAIDs
- pha_opi: Opioids
- pha_relx: Skeletal muscle relaxants
- pha_benz:Benzodiazepines
- pha_antd: Antidepressants
- pha_para: Acetaminophen (paracetamol)
- pha_cort: systemic corticosteroids and
- pha_conv: anticonvulsants

Psychological therapies (including cognitive-behavioural therapies) (psy)

Per prior Cochrane review [10], psychological interventions were classed as any intervention that is designed following a psychological theory of behaviour and behaviour change. Mindfulness meditation, or other forms of meditation, were not, by themselves, considered psychological therapies.

Treatments within Class:

- psy_cbt: cognitive behavioural therapies

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- psy_oth: other psychological therapies

Percutaneous procedures (per)

The following percutaneous procedures were considered:

- Radio frequency denervation: Radiofrequency denervation has been defined [11] as "*a minimally invasive and percutaneous procedure performed under local anaesthesia or light intravenous sedation. Radiofrequency energy is delivered along an insulated needle in contact with the target nerves to denature the nerve*". It was initially developed for the lumbar zygopophyseal joint, and is now applied to denervate other joints in the spine [12].
- Spinal cord stimulation: "*This method was a clinical outgrowth from the well-known gate-control theory for segmental pain suppression. The idea was to apply electric stimulation to the dorsal columns of the spinal cord which are easily accessible and contain large diameter afferent fibers. Thus, stimulating electrodes were applied epidurally to the dorsal aspect of the cord. The gate control theory implied that activation of these coarse fiber systems inhibited transmission of nociceptive information at the segmental level and actually predicted that all types of pain would be equally suppressed*." [13,14]
- Percutaneous multifidus stimulation Percutaneous multifidus stimulation involves "*a stimulating probe is placed into the multifidus muscle via percutaneous procedure, using known anatomical landmarks to target the medial branches of the dorsal rami. Electrical stimulation is applied to target the medial branch of the dorsal ramus after the branch exits the intervertebral foramen prior to innervation of the multifidus and facet joints*."[15]
- Percutaneous rhizolysis, medial bundle branch blocks:
    - Percutaneous rhizolysis (radiofrequency neurotomy), medial bundle branch blocks: "Low-back pain may arise from degenerative changes in the posterior joints of the lumbar spine. These joints are innervated by a branch of the posterior primary ramus, which follows an anatomically constant course. Pain impulses from these joints can be interrupted by coagulating the nerve with a radiofrequency wave, the probe having been placed in the area of the nerve percutaneously." [16]
    - Facet joint medial bundle branch radiofrequency ablation (MBB-RFA) "*involves using energy in the radiofrequency range to perform necrosis of specific nerves (medial branches of the dorsal rami in patients with lumbar facetogenic pain), avoiding the neural transmission of pain. The aim of MBB-RFA is to both provide relief of pain and decrease the possibility of recurrence*". [17,18]

Treatments within Class:

- per_rad: Radio frequency denervation
- per_ssc: Spinal cord stimulation
- per_mfs: Percutaneous multifidus stimulation
- per_rhi: Percutaneous rhizolysis (radiofrequency neurotomy), medial bundle branch blocks
- per_mmb: Facet joint medial bundle branch radiofrequency ablation (MBB-RFA)

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Surgery (sur)

The following types of surgery were included:

- Discectomy (any type): open discectomy, sequestrectomy or aggressive discectomy, microdiscectomy, endoscopic open/percutaneous discectomy, automated open/percutaneous discectomy
- Non fusion stabilization [19]: Graf ligament, Dynesys, interspinous stabilisation devices (e.g., Coflex, Wallis ligament, DIAM), total disc arthroplasty (replacement), facet arthroplasty/facet replacement
- Fusion [19]: anterior, posterior, or circumferential spinal fusion (decompression/discectomy/laminectomy/laminotomy) with/without autologous bone graft harvested from the iliac crest or use of allograft femoral rings stuffed with autologous cancellous bone with/without pedicle screw [20]

Surgery may include indirect/direct decompression [21], decompression with/without instrumentation fusion [21,22] PLIF, ALIF, TLIF, minimally invasive spine surgeries (including laparoscopic ALIF, minimally invasive PLIF, XLIF, OLIF, AxiaLIF).

Treatments within Class:

- sur_dis: Discectomy (any type)
- sur_nstab: Non fusion stabilization
- sur_fus: Fusion
- sur_deco: Decompression/laminectomy/laminotomy without an instrument for foraminal/canal stenosis

Traction (tra)

Traction involves application of a distractive axial force to the spine and trunk for therapeutic effect [23]: "*Mechanical or motorized traction (where the traction is exerted by a motorized pulley), manual traction (in which the traction is exerted by the therapist, using his or her body weight to alter the force and direction of the pull), and auto-traction (where the person controls the traction forces by grasping and pulling bars at the head of the traction table)*" [23] were included as traction. Other forms of traction may include the use of gravity to generate the traction force (e.g., on a tilted table, or hung vertically by the lower extremities).

Treatments within Class:

- tra_mech: Mechanical or motorized traction
- tra_man: manual traction
- tra_auto: auto-traction and use of gravity to generate the traction force

Multidisciplinary (multidisciplinary pain management) (mul)

Multidisciplinary pain management incorporates a number of intervention types, such as education (e.g., mechanisms of chronic pain, anatomy), goal setting, exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment [24–26]. These may be done as individual sessions or as group sessions. If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary pain management. Other studies may have combined individual interventions

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into a multidisciplinary program but did not specifically label it as multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as 'multidisciplinary pain management'. Otherwise, these interventions were classified under 'treatment combinations' (below).

Treatments within Class:

- mul_mdp: Multidisciplinary pain management

Physical therapy (otherwise not falling into specific treatment combination) (pio)
Into this group fall any interventions that are generic 'physiotherapy' or 'physical therapy' treatments, often at the discretion of the clinician, but otherwise not detailed or defined.

Treatments within Class:

- pio_pio: generic physiotherapy or physical therapy treatments

Placebo or sham (pla)
Any intervention defined as a placebo or sham intervention by the study authors, or described as such consistent with previous meta-analysis [27].

Treatments within Class:

- pla_pla: placebo

"Usual care" (e.g., GP Management) (usu)
Intervention deemed 'usual care', including GP management.

Treatments within Class:

- usu_usu: usual care

No treatment (true control) (tru)
No intervention provided, including waitlist control where no treatment is given.

Treatments within Class:

- tru_tru: true control, no intervention
- tru_wait: waiting list control where not treatment is given

Combinations of the above treatments were included and classified according to their primary and secondary treatment components via agreement between the extractors (with adjudication where necessary)

Definition of primary and secondary INT components
The following approach was used to classify primary and secondary intervention components in groups that receive multiple treatments within the same treatment group but did not clearly fall under the multidisciplinary definition above:

1) Pick the primary intervention that contributes to the treatment group: if an intervention comprised >50% of the treatment (per judgement of the extractor), then it was taken as 'primary'. If no intervention component was >50%, then pick the one with the highest proportion.

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In cases that were unclear, the following hierarchy of guiding principles was used:

- Any prior publications (e.g., protocol paper, primary outcome publication) arising from the same study were checked.
- A treatment component that is more thoroughly described could be considered the primary component. For example, if exercise was fully described but advice is labelled as "advice" and not described in similar detail, then exercise was considered as the primary.
- Where a treatment component was mentioned in either the article title or the group subheading was labelled as one of the interventions, then that was taken as the primary intervention component. For example, if the group subheading was called "exercise" but it contained exercise and advice components, then exercise was considered the primary component).
- To split true stalemates, the intervention element mentioned first in the treatment description and/or label was taken to be the primary component (e.g., "exercise and advice" = exercise mentioned first and therefore primary component).
- A minimum threshold to be classified as a primary component was 25%.

2) Secondary component of treatments with multiple components: in some cases, a treatment group may have more than two components, but not fall under the multidisciplinary definition. In this case, the following principles were followed: only ONE secondary intervention component was included, regardless of how many there were. To qualify as being classified as a secondary treatment component, it needed to represent at least 20% of the total intervention (per judgement of the extractor), otherwise the intervention will be classified as having only a primary intervention with no secondary component.

3) If a study arm could not be classified according to the above criteria, then it was treated as a non-included INT (see below). We considered including an additional 'multimodal' category beyond the multidisciplinary group defined above. However, we determined this would be uninformative as it would encompass a heterogeneous range of treatments and thus not provide useful guidance for clinical practice.

Where both primary and secondary intervention components are present we will include these in analyses as combinations of intervention and they will be analysed separately. For example, Physical therapy as a primary component and Massage as a secondary component will be analysed as “Physical therapy + massage”. Due to the approach we have described for classifying primary and secondary components, the order of components may be important, such that we assume that “Physical therapy + massage” is not the same as “Massage + physical therapy”.

Where a secondary intervention component is given in all arms of a study, in addition to the analysis above we will also fit a model in which the study treatments are coded as only the primary intervention in order to test whether assuming additivity of treatment efficacy is reasonable, as this may lead to better connected NMAs with more precise estimates.

Handling of studies that examined non-included INTs

Some studies will examine an INT that is not subject of the current review. In this case, the arms in the study were assessed on a case by case basis.

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- If the 'primary treatment component' of an arm was a non-included INT (e.g., back belts), then that individual arm was not included in extraction and therefore analysis.
- If the 'primary treatment component' was an included INT but the 'secondary treatment component' a non-included INT, then the individual arm was included.
- Pending these decisions, if at least two arm of an individual study could be included, then the study as a whole was included. Otherwise it was excluded.

For example, in the case of a three arm study [28] on "back belt + exercise" vs "exercise" vs "control", the "back belt + exercise" arm was excluded, but the "exercise" and "control" arms were included. Thus the study could also be included.

1 Furlan AD, van Tulder MW, Cherkin D, *et al.* Acupuncture and dry-needling for low back pain. *Cochrane Database Syst Rev* Published Online First: 24 January 2005. doi:10.1002/14651858.CD001351.pub2

2 Engers AJ, Jellema P, Wensing M, *et al.* Individual patient education for low back pain. *Cochrane Database Syst Rev* Published Online First: 23 January 2008. doi:10.1002/14651858.CD004057.pub3

3 Watson T. Key concepts with electrophysical agents. *Phys Ther Rev* 2010;**15**:351–9. doi:10.1179/1743288X10Y.0000000009

4 Oliveira CB, Maher CG, Ferreira ML, *et al.* Epidural corticosteroid injections for lumbosacral radicular pain. *Cochrane Database Syst Rev* Published Online First: 9 April 2020. doi:10.1002/14651858.CD013577

5 Hayden JA, van Tulder MV, Malmivaara A, *et al.* Exercise therapy for treatment of non-specific low back pain. *Cochrane Database Syst Rev* 2005;:77. doi:10.1002/14651858.CD000335.pub2

6 Rubinstein SM, van Middelkoop M, Assendelft WJ, *et al.* Spinal manipulative therapy for chronic low-back pain. *Cochrane Database Syst Rev* Published Online First: 16 February 2011. doi:10.1002/14651858.CD008112.pub2

7 Furlan AD, Giraldo M, Baskwill A, *et al.* Massage for low-back pain. *Cochrane Database Syst Rev* Published Online First: 1 September 2015. doi:10.1002/14651858.CD001929.pub3

8 McKenzie R, May S. *The lumbar spine, volume one: mechanical diagnosis and therapy*. Wellington, N.Z: : Spinal Publications 2003.

9 Surkitt LD, Ford JJ, Hahne AJ, *et al.* Efficacy of Directional Preference Management for Low Back Pain: A Systematic Review. *Phys Ther* 2012;**92**:652–65. doi:10.2522/ptj.20100251

10 Williams AC de C, Fisher E, Hearn L, *et al.* Psychological therapies for the management of chronic pain (excluding headache) in adults. *Cochrane Database Syst Rev* Published Online First: 12 August 2020. doi:10.1002/14651858.CD007407.pub4

11 NICE Guidelines. Chapter 23 Radiofrequency denervation for facet joint pain. In: *Sciatica in Over 16s: Assessment and management*. London: : National Institute for Health and Care Excellence (UK) 2016.

12 Maas ET, Ostelo RW, Niemisto L, *et al.* Radiofrequency denervation for chronic low back pain. *Cochrane Database Syst Rev* Published Online First: 23 October 2015. doi:10.1002/14651858.CD008572.pub2

13 Linderoth B, Foreman RD. Physiology of Spinal Cord Stimulation: Review and Update. *Neuromodulation Technol Neural Interface* 1999;**2**:150–64. doi:10.1046/j.1525-1403.1999.00150.x

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14 Oakley JC, Prager JP. Spinal Cord Stimulation: Mechanisms of Action. *Spine* 2002;**27**:2574–83. doi:10.1097/00007632-200211150-00034
15 Cohen S, Gilmore C, Kapural L, *et al.* Percutaneous Peripheral Nerve Stimulation for Pain Reduction and Improvements in Functional Outcomes in Chronic Low Back Pain. *Mil Med* 2019;**184**:537–41. doi:10.1093/milmed/usy310
16 McCulloch JA. Percutaneous Radiofrequency Lumbar Rhizolysis (Rhizotomy). *Stereotact Funct Neurosurg* 1976;**39**:87–96. doi:10.1159/000102481
17 Shealy CN. Percutaneous radiofrequency denervation of spinal facets: Treatment for chronic back pain and sciatica. *J Neurosurg* 1975;**43**:448–51. doi:10.3171/jns.1975.43.4.0448
18 Akbary K, Kim J-S. Ablation of Medial Bundle Branch Under Spinal Endoscopy. In: Kim J-S, Lee JH, Ahn Y, eds. *Endoscopic Procedures on the Spine*. Singapore: : Springer Singapore 2020. 313–20. doi:10.1007/978-981-10-3905-8_24
19 Baliga S, Treon K, Craig NJA. Low Back Pain: Current Surgical Approaches. *Asian Spine J* 2015;**9**:645. doi:10.4184/asj.2015.9.4.645
20 Christensen FB, Bünger C. Stabilisation surgery for chronic low back pain: indications, surgical procedures, and outcome. *Scand J Rheumatol* 2004;**33**:210–7. doi:10.1080/03009740410005458
21 Watters WC, Bono CM, Gilbert TJ, *et al.* An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spondylolisthesis. *Spine J Off J North Am Spine Soc* 2009;**9**:609–14. doi:10.1016/j.spinee.2009.03.016
22 Kreiner DS, Shaffer WO, Baisden JL, *et al.* An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update). *Spine J Off J North Am Spine Soc* 2013;**13**:734–43. doi:10.1016/j.spinee.2012.11.059
23 Wegner I, Widyahening IS, van Tulder MW, *et al.* Traction for low-back pain with or without sciatica. *Cochrane Database Syst Rev* Published Online First: 19 August 2013. doi:10.1002/14651858.CD003010.pub5
24 Cassidy EL, Atherton RJ, Robertson N, *et al.* Mindfulness, functioning and catastrophizing after multidisciplinary pain management for chronic low back pain: *Pain* 2012;**153**:644–50. doi:10.1016/j.pain.2011.11.027
25 Maruta T, Swanson DW, McHardy MJ. Three year follow-up of patients with chronic pain who were treated in a multidisciplinary pain management center: *Pain* 1990;**41**:47–53. doi:10.1016/0304-3959(90)91108-U
26 Jensen MP, Turner JA, Romano JM. Changes after multidisciplinary pain treatment in patient pain beliefs and coping are associated with concurrent changes in patient functioning: *Pain* 2007;**131**:38–47. doi:10.1016/j.pain.2006.12.007
27 Hrobjartsson A, Gotzsche PC. Is the placebo powerless? Update of a systematic review with 52 new randomized trials comparing placebo with no treatment. *J Intern Med* 2004;**256**:91–100. doi:10.1111/j.1365-2796.2004.01355.x
28 Dalichau S, Scheele K. Auswirkungen elastischer Lumbal-Stützgurte auf den Effekt eines Muskeltrainingsprogrammes für Patienten mit chronischen Rückenschmerzen. *Z Für Orthop Ihre Grenzgeb* 2000;**138**:8–16. doi:10.1055/s-2000-10106

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Supplemental Data B: Search Strategy

Date of database search: 13.11.2019

MEDLINE

Search	Query	Hits
#1	back pain[MeSH Terms] OR low back pain[MeSH Terms] OR back pain*[Title/Abstract] OR lumb* pain[Title/Abstract] OR lumbago[Title/Abstract] OR backache*[Title/Abstract] OR back ache*[Title/Abstract] OR spinal stenosis[Title/Abstract] OR canal stenosis[Title/Abstract] OR lumbar stenosis[Title/Abstract] OR lateral stenosis[Title/Abstract] OR foramin stenosis[Title/Abstract] OR neurogenic claudication[Title/Abstract] OR radiculopathy[Title/Abstract] OR radicular pain[Title/Abstract] OR spondylolisthesis[Title/Abstract] OR spondylosis[Title/Abstract] OR sciatica[Title/Abstract] OR intervertebral disc displacement[Title/Abstract] OR referred pain[Title/Abstract] OR spinal nerve roots[Title/Abstract] OR neurologic signs[Title/Abstract] OR radiat* pain[Title/Abstract] OR radiat* symptoms[Title/Abstract] OR parathesia[Title/Abstract] OR numbness[Title/Abstract]	131336
#2	randomized[Title/Abstract] OR placebo[Title/Abstract] OR randomly[Title/Abstract] OR "drug therapy"[Title/Abstract] OR trial[Title/Abstract] OR groups[Title/Abstract]	2868072
#3	(animals[MeSH] NOT humans[MeSH]))	4659784
#4	#1 AND #2	25960
#5	#4 NOT #3	24928
#6	#5 AND Filters: Randomized Controlled Trial; Clinical Trial; Humans	7237

SPORTDiscus

Search	Query	Hits
#1	(DE "lumbar pain") OR (DE backache) OR (TI ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	11187
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	197381
#4	#1 AND #2	2970

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CINAHL

Search	Query	Hits
#1	(MH low back pain) OR (MH back pain) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	45198
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	889333
#3	MH animals NOT MH human	74138
#4	#1 AND #2	11513
#5	#4 NOT #3	11461
#4	#5 AND Filters: Exclude MEDLINE records; Human; Randomized Controlled Trial	1335

PsycINFO

Search	Query	Hits
#1	(MA low back pain) OR (MA back pain) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	8813
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	1030813
#3	MA animals NOT MA human	196321
#4	#1 AND #2	2829

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#5	#4 NOT #3	2809
#6	#5 AND Filters: Human; Journal Article	2377

EMBASE

Search	Query	Hits
#1	'low back pain'/exp OR 'backache'/exp OR 'back pain*':ab,ti OR 'lumb* pain':ab,ti OR lumbago:ab,ti OR backache*:ab,ti OR 'back ache*':ab,ti OR 'spinal stenosis':ab,ti OR 'canal stenosis':ab,ti OR 'lumbar stenosis':ab,ti OR 'lateral stenosis':ab,ti OR 'foramin stenosis':ab,ti OR 'neurogenic claudication':ab,ti OR radiculopathy:ab,ti OR 'radicular pain':ab,ti OR spondylolisthesis:ab,ti OR spondylosis:ab,ti OR sciatica:ab,ti OR 'intervertebral disc displacement':ab,ti OR 'referred pain':ab,ti OR 'spinal nerve roots':ab,ti OR 'neurologic signs':ab,ti OR 'radiat* pain':ab,ti OR 'radiat* symptoms':ab,ti OR parathesia:ab,ti OR numbness:ab,ti	161402
#2	randomized:ab,ti OR placebo:ab,ti OR randomly:ab,ti OR "drug therapy":ab,ti OR trial:ab,ti OR groups:ab,ti	3957907
#3	'animal'/exp NOT 'human'/exp	5386039
#4	#1 AND #2	33606
#5	#4 NOT #3	32975
#6	#5 AND Filters: Controlled Clinical Trial; Randomized Controlled Trial; Exclude MEDLINE	2627

CENTRAL

Search	Query	Hits
#1	(MeSH descriptor: [back pain] explode all trees) OR (MeSH descriptor: [low back pain] explode all trees) OR ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness):ti,ab,kw	3401
#2	(randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups):ti,ab,kw	1204707
#3	#1 AND #2	2895
#4	(MeSH descriptor: [Animals] explode all trees) NOT (MeSH descriptor: [Humans] explode all trees)	7286
#5	#4 NOT #3	2893
#6	#5 AND Filters: Exclude MEDLINE; Exclude EMBASE; Trials	456

TOTAL from data base searches (with duplicates): 17002

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Date of database search: 24/07/2020

MEDLINE

Search	Query	Hits
#1	back pain[MeSH Terms] OR low back pain[MeSH Terms] OR sciatica[MeSH Terms] OR back pain*[Title/Abstract] OR lumb* pain[Title/Abstract] OR lumbago[Title/Abstract] OR backache*[Title/Abstract] OR back ache*[Title/Abstract] OR spinal stenosis[Title/Abstract] OR canal stenosis[Title/Abstract] OR lumbar stenosis[Title/Abstract] OR lateral stenosis[Title/Abstract] OR foramin stenosis[Title/Abstract] OR neurogenic claudication[Title/Abstract] OR radiculopathy[Title/Abstract] OR radicular pain[Title/Abstract] OR spondylolisthesis[Title/Abstract] OR spondylosis[Title/Abstract] OR sciatica[Title/Abstract] OR intervertebral disc displacement[Title/Abstract] OR referred pain[Title/Abstract] OR spinal nerve roots[Title/Abstract] OR neurologic signs[Title/Abstract] OR radiat* pain[Title/Abstract] OR radiat* symptoms[Title/Abstract] OR paresthesia[Title/Abstract] OR paraesthesia[Title/Abstract] OR numbness[Title/Abstract]	141,803
#2	randomized[Title/Abstract] OR placebo[Title/Abstract] OR randomly[Title/Abstract] OR "drug therapy"[Title/Abstract] OR trial[Title/Abstract] OR groups[Title/Abstract]	2972235
#3	animals[MeSH] NOT humans[MeSH]	4720975
#4	#1 AND #2	28330
#5	#4 NOT #3	27250
#6	#5 AND Filters: Randomized Controlled Trial; Clinical Trial; Humans	9188

SPORTDiscus

Search	Query	Hits
#1	(DE "lumbar pain") OR (DE backache) OR (DE sciatica) OR (TI ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	14,427
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	237964
#4	#1 AND #2	4142

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CINAHL

Search	Query	Hits
#1	(MH low back pain) OR (MH back pain) OR (MH sciatica) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	52162
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	997530
#3	MH animals NOT MH human	79989
#4	#1 AND #2	13351
#5	#4 NOT #3	13289
#4	#5 AND Filters: Exclude MEDLINE records; Human; Randomized Controlled Trial	699

PsycINFO

Search	Query	Hits
#1	(MA low back pain) OR (MA back pain) OR (MA sciatica) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	9726
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	1055873
#3	MA animals NOT MA human	197309
#4	#1 AND #2	3091

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#5	#4 NOT #3	3070
#6	#5 AND Filters: Human; Journal Article	2628

EMBASE

Search	Query	Hits
#1	'low back pain'/exp OR 'backache'/exp OR 'sciatica'/exp OR 'back pain*':ab,ti OR 'lumb* pain':ab,ti OR lumbago:ab,ti OR backache*:ab,ti OR 'back ache*':ab,ti OR 'spinal stenosis':ab,ti OR 'canal stenosis':ab,ti OR 'lumbar stenosis':ab,ti OR 'lateral stenosis':ab,ti OR 'foramin stenosis':ab,ti OR 'neurogenic claudication':ab,ti OR radiculopathy:ab,ti OR 'radicular pain':ab,ti OR spondylolisthesis:ab,ti OR spondylosis:ab,ti OR sciatica:ab,ti OR 'intervertebral disc displacement':ab,ti OR 'referred pain':ab,ti OR 'spinal nerve roots':ab,ti OR 'neurologic signs':ab,ti OR 'radiat* pain':ab,ti OR 'radiat* symptoms':ab,ti OR paresthesia:ab,ti OR paraesthesia:ab,ti OR numbness:ab,ti	176118
#2	randomized:ab,ti OR placebo:ab,ti OR randomly:ab,ti OR "drug therapy":ab,ti OR trial:ab,ti OR groups:ab,ti	4102141
#3	'animal'/exp NOT 'human'/exp	5464750
#4	#1 AND #2	37042
#5	#4 NOT #3	36356
#6	#5 AND Filters: Controlled Clinical Trial; Randomized Controlled Trial; Exclude MEDLINE	2718

CENTRAL

Search	Query	Hits
#1	(MeSH descriptor: [back pain] explode all trees) OR (MeSH descriptor: [low back pain] explode all trees) OR (MeSH descriptor: [sciatica] explode all trees) OR ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness):ti,ab,kw (Word variations have been searched):ti,ab,kw	23060
#2	(randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups):ti,ab,kw	1228587
#3	(MeSH descriptor: [Animals] explode all trees) NOT (MeSH descriptor: [Humans] explode all trees)	606
#4	#1 AND #2	19342
#5	#4 NOT #3	19340
#6	#5 AND Filters: Exclude MEDLINE; Exclude EMBASE; Trials	1258

TOTAL from data base searches (with duplicates): 20633
TOTAL from prior systematic reviews (with duplicates): 1783

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TOTAL from reference lists of 17 relevant Cochrane reviews not included in reviews from last 10 years: 663
Duplicates removed (by Covidence): 3557
TOTAL for screening: 19522

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Supplemental Data C: GRADE Criteria

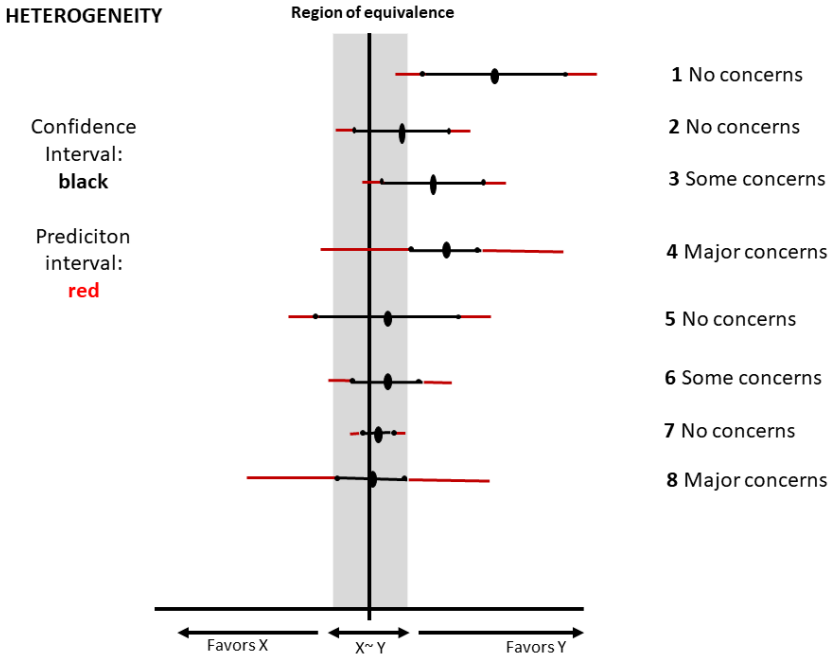
1. Limitations in study design – Cochrane Risk of Bias Tool 1.0

- Selection bias (random sequence generation, allocation concealment, group similarities at baseline);
- Performance bias (blinding of participants and/or healthcare providers);
- Attrition bias (drop outs and intention-to-treat analysis);
- Detection bias (blinding of the outcome assessors and timing of outcome assessment);
- Reporting bias (selective reporting).

We downgraded the quality of the evidence:

- By one level if >50% of participants were from studies with selection bias **and** performance bias.
- Inadequate randomization and lack of blinding may lead to an exaggeration of the intervention effect estimates [1–3].

Unexplained heterogeneity or inconsistency of results



- Pre-defined area/range of equivalence: We define a range of equivalence of SMD -0.5 to 0.5 [4].
- Downgrade two levels if there is a major concern and one level if there are some concerns.
- If there are very few trials, the amount of heterogeneity is poorly estimated and prediction intervals are unreliable, we will downgrade based on reference priors [5].

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Indirectness

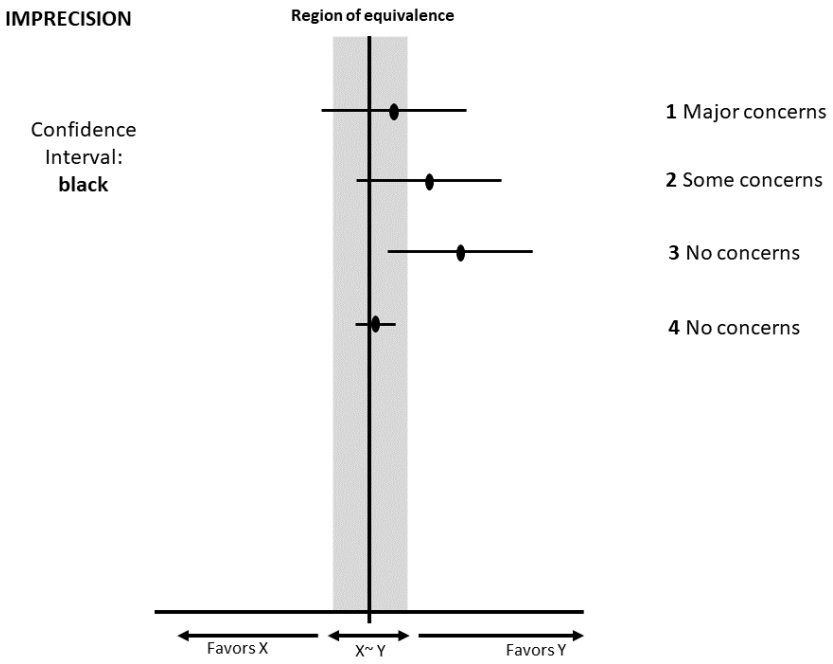
Domain (original question asked)	Description (evidence found and included, including evidence from other studies) – consider the domains of study design and study limitation, inconsistency, imprecision and publication bias	Judgment – is the evidence sufficiently direct?			
Population:		Yes	Probably yes	Probably no	No
Intervention:		Yes	Probably yes	Probably no	No
Comparator:		Yes	Probably yes	Probably no	No
Direct comparison:		Yes	Probably yes	Probably no	No
Outcome:		Yes	Probably yes	Probably no	No
Final judgement about indirectness across domains:	No indirectness ? => No downgrade. Serious indirectness ? => Downgrade one level. Very serious indirectness ? => Downgrade two levels.				

Two components for network meta-analysis:

- similarity of studies in the analysis to the target question (PICO)
- similarity of the studies in the analysis to each other (relates to transitivity assumption)

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Imprecision



- Downgrade two levels if there is a major concern and one level if there are some concerns.

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Publication bias [6]

Reporting bias may be suspected when the following occur:

- Prior documented evidence of reporting bias in trials in the field.
- meta-analysis is based on a small number of new studies, typically positive findings (e.g. new drugs may have positive findings early and later the true effect size becomes apparent).
- Industry-funded trials dominate
- Known unpublished data from grey literature not included.

Reporting bias is considered to not be present in the following situations:

- Analytical methods indicate the findings from small are similar to those in large/published studies
- Findings from unpublished studies agree with published studies.
- Prospective trial registration, protocol publication and/or clinical trial registries are used extensively in the field and do not indicate important discrepancies with published reports.

⇨ Downgrade one level if publication bias is suspected.

Criteria specific to NMA:

- Do **not consider imprecision** when rating the direct and indirect estimates to inform the rating of NMA estimates [7].
- **No need to rate the indirect evidence** when the certainty of the direct evidence is high and the contribution of the direct evidence to the network estimate is at least as great as that of the indirect evidence.

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1 Savović J, Turner RM, Mawdsley D, *et al.* Association Between Risk-of-Bias Assessments and Results of Randomized Trials in Cochrane Reviews: The ROBES Meta-Epidemiologic Study. *Am J Epidemiol* 2018;**187**:1113–22. doi:10/gdgm64

2 Armijo-Olivo S, Fuentes J, da Costa BR, *et al.* Blinding in Physical Therapy Trials and Its Association with Treatment Effects: A Meta-epidemiological Study. *Am J Phys Med Rehabil* 2017;**96**:34–44. doi:10/gmhbgv

3 Armijo-Olivo S, Saltaji H, Costa BR da, *et al.* What is the influence of randomisation sequence generation and allocation concealment on treatment effects of physical therapy trials? A meta-epidemiological study. *BMJ Open* 2015;**5**:e008562. doi:10/gb5db8

4 Norman GR, Sloan JA, Wyrwich KW. Interpretation of changes in health-related quality of life: the remarkable universality of half a standard deviation. *Med Care* 2003;**41**:582–92. doi:10/dhstrr

5 Rhodes KM, Turner RM, Higgins JPT. Predictive distributions were developed for the extent of heterogeneity in meta-analyses of continuous outcome data. *J Clin Epidemiol* 2015;**68**:52–60. doi:10/f6thkp

6 Nikolakopoulou A, Higgins JPT, Papakonstantinou T, *et al.* CINeMA: An approach for assessing confidence in the results of a network meta-analysis. *PLOS Med* 2020;**17**:e1003082. doi:10/ggthw8

7 Brignardello-Petersen R, Bonner A, Alexander PE, *et al.* Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis. *J Clin Epidemiol* 2018;**93**:36–44. doi:10/gcz4pg

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-P reporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a - not an update

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	2
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a - not an amended version
Support			
Sources	#5a	Indicate sources of financial or other support for the review	2
Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a - no sponsor
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a - no sponsor
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	5-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6–7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6; 9–10
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	9; (Supplemental data B, 36-42)
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through	9–10

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		each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	10–11
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	10
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	11–12
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	13-18
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	14–15

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Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	14-15
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	12

Notes:

- 1b: n/a - not an update
- 4: n/a - not an amended version
- 5b: n/a - no sponsor
- 5c: n/a - no sponsor
- 10: 27-33 (Supplemental data B) The PRISMA-P elaboration and explanation paper is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 17. March 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5–7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	9
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	9; Supplemental Data B, 36–42
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	9–10
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	10–11
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	10–11
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	10; 13
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	11–12
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	13
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	13-18
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	13; 17
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	13-18
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	13–15
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	17
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	13; 17
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	12-13; 17-18

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	12-13
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	
	23b	Discuss any limitations of the evidence included in the review.	
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	
Competing interests	26	Declare any competing interests of review authors.	
Availability of data, code and	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
other materials			

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Network meta-analysis for comparative effectiveness of treatments for chronic low back pain disorders: Systematic review protocol

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Manuscript ID	bmjopen-2021-057112.R1
Article Type:	Protocol
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Complete List of Authors:	Belavy, Daniel; Hochschule fur Gesundheit, Physiotherapy Diwan, Ashish; St. George Hospital, Department of Orthopaedic Surgery, Spine Service Ford, Jon; Advance Healthcare; La Trobe University College of Science Health and Engineering, Low Back Research Team Miller, Clint; Deakin University Faculty of Health, Institute for Physical Activity and Nutrition (IPAN), School of Exercise and Nutrition Sciences Hahne, Andrew; La Trobe University College of Science Health and Engineering, Low Back Research Team Mundell, Niamh ; Deakin University Tagliaferri, Scott ; Deakin University Bowe, Steven; Deakin University Faculty of Health, Biostatistics Unit Pedder, Hugo ; University of Bristol; ; UK , Bristol Medical School Saueressig, Tobias ; Physio Meets Science GmbH, Zhao, Xiaohui; Xi'an University of Architecture & Technology Chen, Xiaolong; University of New South Wales, Department of Orthopaedic Surgery, Spine Service Balasundaram, Arun Prasad; La Trobe University College of Science Health and Engineering, Low Back Research Team Arora, Nitin Kumar; Jamia Millia Islamia, Centre for Physiotherapy and Rehabilitation sciences; Hochschule fur Gesundheit, Physiotherapy Owen, Patrick; Deakin University, Institute for Physical Activity and Nutrition
Primary Subject Heading:	Rehabilitation medicine
Secondary Subject Heading:	Sports and exercise medicine, Surgery
Keywords:	Spine < ORTHOPAEDIC & TRAUMA SURGERY, Back pain < ORTHOPAEDIC & TRAUMA SURGERY, REHABILITATION MEDICINE, PAIN MANAGEMENT

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Network meta-analysis for comparative effectiveness of treatments for chronic low back pain disorders: systematic review protocol

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Short title: Treatments for chronic back disorders: NMA protocol

Word count: 4372 (excluding abstract, references, text in figures and supplementary file)

Key words: Spine; Lower back pain; Sciatica; Back pain; Rehabilitation; Physical therapy; Surgery; Psychotherapy; Analgesia

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ABSTRACT

Introduction: Chronic low back pain disorders (CLBDs) present a substantial societal burden; however, optimal treatment remains debated. To date, pairwise and network meta-analyses have evaluated individual treatment modes, yet a comparison of a wide range of common treatments is required to evaluate their relative effectiveness. Using network meta-analysis, we aim to evaluate the effectiveness of treatments (acupuncture, education or advice, electrophysical agents, exercise, manual therapies/manipulation, massage, the McKenzie method, pharmacotherapy, psychological therapies, surgery, epidural injections, percutaneous treatments, traction, physical therapy, multidisciplinary pain management, placebo, "usual care" and/or no treatment) on pain intensity, disability and/or mental health in patients with CLBDs.

Methods and analysis: Six electronic databases and reference lists of 285 prior systematic reviews were searched. Eligible studies will be randomised controlled/clinical trials (including cross-over and cluster designs) that examine individual treatments or treatment combinations in adult patients with CLBDs. Studies must be published in English, German or Chinese as a full-journal publication in a peer-reviewed journal. A narrative approach will be used to synthesise and report qualitative and quantitative data, and, where feasible, network meta-analyses will be performed. Reporting of the review will be informed by Preferred Reporting Items for Systematic Review and Meta-Analysis guidance (PRISMA), including the network meta-analysis extension (PRISMA-NMA). The Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach for network meta-analysis will be implemented for assessing the quality of the findings.

Ethics and dissemination: Ethical approval is not required for this systematic review of the published data. Findings will be disseminated via peer-reviewed publication.

Registration: PROSPERO registration number CRD42020182039.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- This study will enable comparison of a wide variety of treatments for chronic low back disorders via network meta-analysis.
- Our study will provide evidence that can be applied in clinical practice and in low back pain management guidelines.
- The quality of evidence will be assessed via the Grading of Recommendations Assessment, Development and Evaluation (GRADE).
- We will address the potential limitation of heterogeneous pathologies being combined into one population by performing sub-group analyses.
- Baseline pain and disability are known to be predictive of outcome and we will account for this in the analysis.

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INTRODUCTION

Low back pain is the greatest cause of disability and lost productivity world-wide [1]. In developed regions, such as the United States of America, Japan, Europe and Australia, the disease generates substantial financial costs [2]. For example, healthcare expenditure is in excess of A$5 billion per year in Australia [3] and US$100 billion per year in the United States of America [3]. The majority of acute cases of back pain resolve without specific intervention, [4] yet chronic low back pain disorders (CLBDs; i.e. >12 weeks duration) generate the greatest proportion of economic burden [5] and affect 20.1 ± 9.8 % of the population worldwide [6]. To reduce the global burden of disease of CLBDs, identifying and implementing the most effective treatment is an urgent priority [7].

To date, pairwise meta-analyses have typically been used to evaluate individual treatment modes for CLBDs [8]. Current recommendations include education, exercise, manual therapy, psychotherapy and multidisciplinary interventions [8,9]. A comparison of a wide range of common treatments and their relative effectiveness for CLBDs is yet to be performed. This evidence would inform management guidelines and clinical decision making. These data would also increase the likelihood that patients receive the most efficacious treatment and/or avoid therapies with similar effectiveness but greater harms. Collectively, this would reduce financial burden at the societal level, as well as improve patient outcomes at the individual level.

Network meta-analysis (NMA) permits the ranking of a series of interventions as comparably more or less effective [10,11]. NMA can incorporate data on multiple treatments simultaneously from randomised controlled trials (RCTs) that do not have similar comparator groups by synthesising direct and indirect evidence from a ‘network’ of studies [11–13]. This

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overcomes a key limitation for pairwise meta-analysis, and allows RCTs that do not have a non- or minimal-treatment control group to be included in the analysis [14]. NMA has been used to examine the relative effectiveness of exercise training modalities in non-specific chronic low back pain [15], exercise and education for back pain prevention [16], treatments for lumbar disc herniation [17] and medication for sciatica [18]. However, this approach has not been considered simultaneously for a wide range of common treatments of CLBDs. In this study, we will examine CLBDs, encompassing radicular syndromes and non-specific low back pain [19]. Our primary aim is to determine the relative effectiveness of a variety of common treatments for CLBDs via NMA.

METHODS

This systematic review will be conducted in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [20] and the PRISMA extension for network meta-analyses (PRISMA-NMA) [21]. Covidence (https://www.covidence.org) will be used for article screening and data extraction. This systematic review was prospectively registered on PROSPERO (submitted 24th April 2020; registration number CRD42020182039) prior to initiating data extraction. We will use the PRISMA-P checklist when writing our report [22].

Eligibility criteria

For inclusion, studies will be required to be full peer-reviewed publications (i.e., grey literature including theses and conference abstracts will be excluded) in English, German or Chinese. A meta-epidemiological study by Nussbaumer-Streit et al. [23] found that when non-English

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studies were excluded from systematic reviews of clinical interventions, this had little impact on study conclusions. Furthermore, Cochrane guidelines [24] are ambivalent on the inclusion of non-English language articles and the potential for introduction of bias in reviews. Prior work has suggested that inclusion or exclusion of non-English articles does not influence the effect estimates, but may narrow confidence intervals [25]. We pragmatically chose to include articles in languages in which the author team were fluent. All other inclusion criteria followed the Participants, Interventions, Comparators, Outcomes and Study design (PICOS) framework [21].

Population: Adults (≥ 18 years) with CLBDs. Chronic is defined as pain lasting 12 weeks or more [26]. Since not all studies are consistent in their reporting of pain duration, we will use the following approach: if a study defines it collectively as "chronic", then it will be included. Failing this, if the inclusion criteria of the study are minimum of 12 weeks pain duration or if the median or mean reported duration of pain at baseline in participants is 12 weeks or more, then the study will be included. Recurrent pain (i.e., <12 weeks duration of symptoms and pain-free period of at least 6 months [27]) is excluded. Low back disorder is defined as back pain with or without leg pain where there are no specific spinal pathologies (i.e., vertebral fracture, malignancy, spinal infection, axial spondyloarthritis, cauda equina syndrome [19]). Spondylolisthesis, spondylosis, disc herniation, disc degeneration, scoliosis, deformity (e.g., hemivertebrae) and radicular syndromes (e.g., radicular pain [leg pain or sciatica], radiculopathy, spinal stenosis) are included [19]. “Failed back surgery syndrome” is included as this is not a specific disease [28]. If a study only examines post-surgical pain (e.g., a comparison of management for immediate post-surgical pain as an RCT), we will consider this iatrogenic pain and the study will be excluded.

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Interventions and comparators: The treatment types to be included were determined by the current clinical practice guideline from the American College of Physicians [29] and by the review areas of the Cochrane Back and Neck Group [30]. A detailed list is included in Supplemental Data A; however, in brief, we examined acupuncture, education or advice, electrotherapy (including heat and ice electrotherapeutic modalities applied non-invasively), epidural injections, exercise training, manual therapies/manipulation, massage, the McKenzie method, pharmacotherapy, psychological therapies, percutaneous procedures, surgery, traction, physical therapy (otherwise not falling into specific treatment combination), placebo, multidisciplinary pain management, usual care (e.g., general practitioner management), no treatment (true control). Treatment combinations will be considered pending data availability and defined according to their component parts (see Supplemental Data A for details) for primary and secondary treatment components. Pending articles included in the review, further sub-group classifications will be considered.

Outcomes: Pain intensity (e.g., VAS, NRS, McGill Pain Questionnaire, or Box scale, other quantitative pain measures), disability (e.g., ODI, RMDQ), mental health (e.g. SF-36 MH subscale, depression, anxiety). Adverse events, participant drop-outs and funding sources will be extracted from the included articles.

Study design: Randomised controlled trials, randomised clinical trials, randomised controlled cluster trials, or randomised cross-over trials will be included.

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Search strategy

Six databases (MEDLINE, SPORTDiscus, CINAHL, PsycINFO, EMBASE, CENTRAL) were searched with no restriction on publication dates. The search was initially performed from inception to 14.11.2019 and then was updated on 24.07.2020. Search terms were to find articles on (1) low back disorders and (2) randomised controlled trials (Supplemental Data B). Low back disorder terms included those recommended by the Cochrane Back and Neck review group [31] for non-specific back pain and radicular syndromes [19]. The search terms for identifying RCTs were modelled on Cochrane sensitivity-maximising and precision-maximising search terms to be consistent across databases. Prior systematic reviews in English of any kind of treatment for chronic low back disorders in the last 10 years were screened via a search (January 1990 to July 2019) of MEDLINE, SPORTDiscus, CINAHL, PsycINFO, EMBASE and CENTRAL. Collectively, 285 such systematic reviews were identified. The complete reference lists of these reviews were collated and then screened to remove non-RCTs. Subsequently, 1783 additional references were identified, and after uploading to Covidence, 1008 duplicates were removed, leaving 775 new titles/abstracts. Furthermore, the reference lists of 17 relevant Cochrane reviews not published between January 1990 and July 2019 were screened: 269 additional references were added after discarding 394 duplicates. Following removal of duplicates, a total of 19522 articles remained for screening.

Study selection

For each record, two independent assessors will screen the studies against the predetermined inclusion/exclusion criteria. Disagreements that cannot be resolved amongst the assessors will be addressed by an adjudicator. If unsure, the adjudicator will discuss with the broader study

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team. If still unsure, the study authors will be contacted for clarity. The process for determining study inclusion/exclusion is shown in Figure 1.

Data extraction

For each record, two independent assessors will extract the data. Disagreements that cannot be resolved amongst the assessors will be addressed by an adjudicator. Relevant information pertaining to publication metadata (i.e., author, title, year, journal), study design (e.g., two-arm or multi-arm parallel trial), number of participants, participant characteristics (e.g., age and sex), interventions considered, and outcome measures (pain, disability, mental health, adverse events and funding sources) will be extracted by two independent assessors. Extracted outcome data (pain, disability, mental health) will be pre- and post-intervention mean and standard deviation (SD). When available, data will be extracted for the following time-points: immediate (<1d) effect of treatment, short-term (≥1d but <3mo), intermediate-term (≥3 but <12mo), long-term (≥12mo). Primary and any secondary intervention components will be labelled as per the protocol described in Supplemental Data A.

Data presented as medians or alternate measures of spread will be converted to mean and SD using established formulae [32]. When only figures are presented (rather than numerical data within text), data will be extracted using ImageJ (https://imagej.nih.gov/ij/) to measure the length (in pixels) of the axes to calibrate, and then the length in pixels of the data points of interest [33]. When it is not possible to extract the required data, this information will be requested from the authors at a minimum of three times over a four-week period. Prior to commencing data extraction, this method will be piloted on 30 studies chosen at random. All discrepancies will be referred to an adjudicator.

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Due to the volume of potentially included articles, for each study, information on the population (type of low back pain [non-specific or radicular pain], and sub-population [e.g., 'non-specific', 'low back pain not otherwise stated', 'disc degeneration', 'spondylolisthesis', 'spinal stenosis', 'radiculopathy', 'radicular pain']) and intervention/comparator (intervention duration, free text entry of description of interventions, study-arm labels, primary and secondary intervention classifications (if relevant); see Supplemental Data A) will be extracted first. Then, studies that examine different treatment classes (e.g., exercise versus control, psychological therapies versus exercise, or surgery versus percutaneous therapies; see Supplemental Data A) will be included in subsequent extraction and the remaining studies excluded. This approach will be undertaken because our primary research question concerns different classes of treatments; hence, studies that compare the same class of treatment (e.g., exercise versus exercise, or surgery versus surgery) are less informative for this question.

Risk of bias

Two independent assessors will use the Cochrane Collaboration Risk of Bias [34] to examine potential selection bias (random sequence generation and allocation concealment), performance bias (blinding of patients and personnel), detection bias (blinding of outcome assessment), attrition bias (incomplete outcome data), reporting bias (selective outcome reporting) and other biases. Cluster randomised trials will be assessed as recommended by the Cochrane Collaboration [35]. The revised version of the risk of bias tool [36] will not be used as it was, at initiation of the project, not yet recommended by the Cochrane Collaboration. For each source of bias, studies will be classified as having a low, high or unclear (if reporting was

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not sufficient to assess a particular domain) risk. All discrepancies will be referred to an adjudicator.

Two independent assessors will use the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach for network meta-analysis for assessing the quality of the evidence (Supplemental Data C). We will use a range of equivalence of standardised mean difference (SMD), from -0.5 to 0.5, to evaluate imprecision and inconsistency [37]. Publication bias will be assessed via statistical and non-statistical methods [38]. Indirectness will be judged using Schünemann's approach [39]. Risk of bias will be downgraded by one level if >50% of participants were from studies with selection bias and performance bias. This criterion was selected because inadequate randomization and lack of blinding may lead to an exaggeration of the intervention effect estimates [40–42]. For the categories 'imprecision' and 'inconsistency', we will downgrade by one level if there are some concerns and two levels if there are major concerns. Indirectness will be downgraded by one level if deemed serious and two levels if deemed very serious. We will downgrade one level if publication bias is suspected. The GRADE approach [43,44] will also be used to assess the quality of the evidence of pair-wise comparisons. All discrepancies will be referred to an adjudicator.

Statistical analyses

When studies are reverse scaled (i.e., higher values indicated better outcomes rather than lower values), the mean in each group will be multiplied by -1 as recommended in the Cochrane Handbook. As all of the outcomes of interest will be continuous or ordinal, but could be measured on different scales, SMD will be used as the effect estimates [45]. A

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minimum of 50 participants will be required per class of treatment for it to be included in meta-analysis. We have limited the number of participants to try to limit the impacts of small study effects on the results of any particular class [45]. Furthermore, because we are conducting an analysis of Standardised Mean Differences (SMD), small study effects are likely to be exacerbated as both the mean and the standard deviations are likely to be estimated with greater variability in small studies, and for SMD both of these contribute to the treatment effect. To further investigate our choice of SMD as an effect measure, we will conduct sensitivity analyses with internal reference baseline SDs for each scale [46].

Where a study does not report data in a form where the SD can be extracted or calculated [32], and authors are not able to fulfil data requests, SDs will be imputed and their impact evaluated in sensitivity analyses. To impute missing SDs, we will perform a regression of log(SD) on log(mean) in studies reporting SD following the approach of Marinho et al. 2003, adjusting for measurement scale and follow-up time [47]. We will then use this regression model to predict SDs that are missing.

Cluster randomised trials will be included in the analysis as per Cochrane guidance. Sensitivity analysis will be conducted in pairwise analyses with a range of different ICCs to check the robustness of the results. [48] For crossover trial designs, we will include the estimated relative treatment effect from the study where possible, where the authors have tested for carryover effects and found no evidence of this. Where this is not the case, we will only include the first period of the crossover trial. In time-course Model-Based Network Meta Analyses (MBNMA), only the inclusion of the first time-period will be possible.

Network Meta-Analysis

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Bayesian network meta-analysis (NMA) will be performed at discrete time-points (immediate (<1d) effect of treatment, short-term (≥1d but <3mo), intermediate-term (≥3 but <12mo), and long-term (≥12mo)) using the R (r-project.org) package multinma [49]. Time-course MBNMA will be conducted using the R package MBNMAtime [50,51]. This package enables the incorporation of multiple time-points per study in Bayesian NMA to inform estimates of effect size over time. Network connectivity will be explored via network plots. Network plots help to visualise how the evidence in the network is connected and allow identification of which studies compare which treatments. This aids in understanding which treatment effects can be estimated. The time-course relationship will be examined by a time plot, which is a plot of the raw study responses over time. Time plots help to elucidate the underlying time course of the treatment effects and help to identify which statistical time model is appropriate.

Where data allow and where there is a plausible clinical reason for doing so, treatment effects will be assumed to be common or exchangeable within a class. This allows for treatments to be nested within a class, which relaxes assumptions regarding the similarity of interventions whilst improving network connectivity [13]. We will use the deviance information criterion (DIC) to compare the different models (common/exchangeable class effect models, time-course models) to assess their parsimony [52].

For standard NMA models we will rank the relative effects of each treatment/class, and for time-course MBNMA models we will rank the relative effects of each treatment/class for each time-course parameter. We will also rank the full area under the time-course function for each treatment/class at 0–3 months, 0–6 months and 0–12 months. Cumulative rankograms will be plotted; these show the range of rankings of different treatments/classes for each ranked parameter. Sensitivity of model results to the choice of prior distributions will be investigated.

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Assessing key assumptions of pairwise and network meta-analysis

The authors recommended a strong and rigorous focus on the evaluation of the similarity and homogeneity assumptions.

Assessment of similarity and homogeneity assumptions

A qualitative assessment of the clinical similarity of the different populations and treatments will be performed by important variables such as baseline pain intensity, baseline disability and pain duration. Between-study SD will be estimated and reported from random effects models, and the impacts of subgrouping or meta-regression on this will be examined. Pair-wise meta-analysis of data will be synthesised via SMDs with accompanying 95% confidence intervals using a frequentist random effects model with a restricted maximum likelihood estimator for the between study variance Tau^2. These analyses will be carried out with the R package "metafor" [53]. Visual inspection of the forest plots, statistical estimates of heterogeneity (I^2, Tau) and 95% prediction intervals will be used to assess the validity of homogeneity assumptions. Small study effects and publication bias will be assessed for each pairwise comparison by visual inspection of the contour-enhanced funnel plot. Outlier and influential study analysis will be performed with metafor for pairwise meta-analyses to further detect potential heterogeneity [54]. Meta-regression with potential effect modifiers (pre-intervention pain severity and disability, baseline psychological conditions, presence of cointerventions and type of low back pain); [55–57] will be used to further check for potential heterogeneity among the pairwise comparisons [58].

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In the presence of effect modification in pairwise comparisons (identified using meta-regression), we will also fit network meta-regression with these potential effect modifiers for NMAs conducted at each time-point using the package multinma [49].

Consistency assumptions

For the Bayesian approach, consistency assumptions will be first checked via an unrelated mean effects (UME) model which does not assume consistency [59]. The UME model only synthesises direct relative effects between each arm in a study and the study reference treatment. If the consistency assumption holds then the results from the UME and NMA models will be similar. Changes in between-study SD or residual deviance are also suggestive of inconsistency. If comparison between UME and NMA models is suggestive of inconsistency, node-splitting will be performed [60]. In node-splitting, network contrasts are split into direct and indirect evidence contributions, which can then be compared to examine their similarity.

Additional assumptions required for analysis of time-course data

Given that data will be reported at different follow-up times in different studies, information is unlikely to be available for all treatments at all time-points of interest. For this reason, additional assumptions regarding specific parameters for treatments/classes may be required. For example, in the case of a treatment for which information is only available at shorter follow-up times, explicit assumptions regarding its long-term efficacy will be required. The treatment's long-term efficacy could be assumed to be the same as (or similar to) that of another treatment in the network that might have a similar mechanism of action (e.g., within the same class), for which long-term data is available. Alternatively, it could be assigned a specific value or an informative prior as determined by clinical expertise. In such an example, long-term

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results for this treatment will therefore be sensitive to these assumptions, and results will be interpreted accordingly [51]. Assumptions made in this way will be clearly stated and justified.

Subgroup and sensitivity analyses

Pending data availability, we will perform subgroup analyses to explore whether inconsistency/heterogeneity and group differences in the outcomes are influenced by type of low back disorder (e.g., non-specific chronic low back pain, radicular syndrome), type of treatment (e.g., surgical, pharmacological) or by exclusion of the multidisciplinary node and the physical therapy (otherwise not falling into specific treatment combination) node from analyses. The treatment node may be a source of significant heterogeneity/inconsistency for the overall NMA due to the variability of this treatment definition compared to other interventions. Subgroup analysis focussing on key participant or study characteristics can produce smaller, more homogenous networks and can be a good strategy to analyse inconsistency/heterogeneity with fewer assumptions and pitfalls then NMA meta-regression [61]. If we are unable to identify the source of inconsistency, we will highlight that this limits the usefulness of the analysis for drawing meaningful conclusions in such a heterogeneous population.

Further, pending data-availability, we will consider the following sensitivity analyses

- Excluding studies with imputed missing standard deviations and imputed medians.
- Study sample size: impact of studies including less than 20 participants in all study-arms.
- Drop out numbers and handling of dropouts within studies: the impact of the proportion of dropouts (if reported) and the kind of analysis individual studies performed (e.g.,

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analysing all participants using imputation of missing data vs. analysing complete cases only).

- Comparison of class effect models to a model with fully independent treatment effects that assume no within-class similarity, to assess the statistical validity of class assumptions.
- Secondary treatment components (see Supplemental Data A): the impact of treatment combinations where secondary classes of treatment are present in all arms will be considered by fitting models that incorporate combinations as different nodes in the network. This can be used to assess the assumption of additivity of combined treatments. We will also investigate the impact of ordering of primary/secondary treatment components by fitting a model in which the order is ignored (e.g. "Physical therapy + massage" assumed to be equivalent to "Massage + physical therapy")
- Secondary treatment components (see Supplemental Data A): the impact on effect estimates of when secondary treatments are included will be assessed via a sensitivity analysis excluding those interventions with a secondary treatment component.
- As some osteopathic interventions may include visceral techniques not declared in the original methods of the study, the impact of removing this from the manual therapy node will be examined.
- Excluding unclear generic nodes (e.g., physical therapy otherwise not falling into specific treatment combination)
- Risk of bias: To examine the influence of specific studies/comparisons on the treatment rankings we will conduct a threshold analysis where possible [51] using the R package nmathresh.
- Choice of SMD as an effect measure by using internal reference baseline SDs for analysis. [46]

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DISCUSSION

This NMA will determine the relative effectiveness of a variety of common treatments for CLBDs. Conducting NMA on this topic constitutes a shift towards the highest level of medical evidence [62]. Our NMA has a much broader scope than prior work, such as that concerned solely with pharmacotherapy [63–66], exercise training [15,67,68], traditional Chinese medicine [69], or psychotherapy [70]. Moreover, the broad inclusion criteria and number of interventions considered in our NMA will result in a greater number of included interventions than previous broad NMAs that examined non-pharmacotherapy [71] and surgery-based interventions [72], which included 31 and 12 interventions, respectively. The breadth of our NMA is important given that CLBDs are inherently heterogenous, yet progenitors do not influence decision making regarding treatment sought [73]. For this reason, CLBDs (excluding specific causes) are commonly treated in line with generic clinical guidelines [74]. This underpins the importance of our NMA, as these guidelines do not distinguish whether one treatment is superior to another for this collective of patients with chronic pain. Given the lack of evidence that treatment efficacy differs by underlying pain progenitor, we believe it is reasonable to assume exchangeability of these studies and transitivity within the network in terms of population. Other than recent suggestions that machine learning [75] may one day identify evidence-based sub-groups that respond 'better' to specific treatments, we surmise that our NMA will markedly contribute to overcoming current limitations in the management of CLBDs pertaining to treatment decision making.

To our knowledge, there is only one other NMA currently being conducted with a similar scope to our protocol [76]. Our NMA overcomes several cardinal limitations of this protocol: (1) we consider CLBD, rather than solely non-specific low back pain; (2) we consider additional languages for article inclusion, rather than English only; and (3) our treatment classification is

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more nuanced, rather than simplistic (e.g., the other protocol typically considers two types of treatment within a particular class). Of note, we registered our systematic review prior to publication of this other protocol, and it is unclear when their work is due to be published.

Despite the many strengths of our proposed NMA, we would be remiss not to acknowledge potential limitations. First, due to the inclusion of radicular syndromes in the patient population, it might be necessary to analyse this population in different networks/subsets because the presence of this may be an effect modifier [77] and lead to intransitivity. Second, we do not consider multicomponent interventions in our statistical model, which might have an impact on the estimates [78,79]. By ignoring additional treatment components given in both arms of included studies, we assume additivity of different treatment components. While we will investigate the effects of this (see *Sensitivity Analyses*), fully accounting for it by modelling all combinations of treatments as separate interventions is likely to lead to disconnected networks of evidence, which poses its own problem for evidence synthesis and decision making [80]. Third, while we propose a variety of subgroup analyses to investigate the impact of effect modification, potential effect modifiers may be poorly reported in many studies. However, there is no clear evidence of important effect modification in CLBD to date. As pointed out in the recent Lancet Low Back Pain Series [8], relative treatment efficacy for different kinds of interventions appears (to date) to be surprisingly similar. Fourth, usual care may vary between included studies (e.g., authors' stance on whether or not usual analgesic pharmacotherapy was permitted), yet given few studies in the CLBD field employ methods of strict observation, we surmise that the majority, if not all, of existing studies are inherently at risk of this form of bias. Finally, as with all meta-analyses, dealing with co-interventions has implicit complexities. Our decision to consider interventions that combine multiple forms of interventions of interest may impede our capacity to differentiate the effects of one individual treatment. However, we

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contend that this approach allows for the inclusion of more trials that, when compared to a strict approach that excluded any interventions with co-intervention, reflects more realistically the realities of clinical practice. This, in our view, leads to less potential bias (e.g. inclusion of studies that simply failed to report co-interventions) and greater confidence in our effect estimates.

In conclusion, the current project will enable a significant advance in synthesising knowledge on the comparative effectiveness of a wide variety of treatments for chronic low back disorders. This has, to date, not been performed and will inform patient management and clinical practice guidelines.

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References

1 Vos T, Flaxman AD, Naghavi M, *et al.* Years lived with disability (YLDs) for 1160 sequelae of 289 diseases and injuries 1990-2010: a systematic analysis for the Global Burden of Disease Study 2010. *Lancet Lond Engl* 2012;**380**:2163–96. doi:10.1016/S0140-6736(12)61729-2

2 Dagenais S, Caro J, Haldeman S. A systematic review of low back pain cost of illness studies in the United States and internationally. *Spine J* 2008;**8**:8–20. doi:10.1016/j.spinee.2007.10.005

3 Martin BI, Deyo RA, Mirza SK, *et al.* Expenditures and health status among adults with back and neck problems. *JAMA* 2008;**299**:656–64. doi:10.1001/jama.299.6.656

4 da C Menezes Costa L, Maher CG, Hancock MJ, *et al.* The prognosis of acute and persistent low-back pain: a meta-analysis. *CMAJ* 2012;**184**:E613-624. doi:10.1503/cmaj.111271

5 Maetzel A, Li L. The economic burden of low back pain: a review of studies published between 1996 and 2001. *Best Pract Res Clin Rheumatol* 2002;**16**:23–30. doi:10.1053/berh.2001.0204

6 Hoy D, Bain C, Williams G, *et al.* A systematic review of the global prevalence of low back pain. *Arthritis Rheum* 2012;**64**:2028–37. doi:10/fzbtrz

7 Hartvigsen J, Hancock MJ, Kongsted A, *et al.* What low back pain is and why we need to pay attention. *The Lancet* 2018;**391**:2356–67. doi:10/gdnd74

8 Maher C, Underwood M, Buchbinder R. Non-specific low back pain. *The Lancet* 2017;**389**:736–47. doi:10/gcpzvb

9 Almeida M, Saragiotto B, Richards B, *et al.* Primary care management of non-specific low back pain: key messages from recent clinical guidelines. *Med J Aust* 2018;**208**:272–5. doi:10.5694/mja17.01152

10 Shim S, Yoon B-H, Shin I-S, *et al.* Network meta-analysis: application and practice using Stata. *Epidemiol Health* 2017;**39**:e2017047. doi:10.4178/epih.e2017047

11 Mills EJ, Thorlund K, Ioannidis JPA. Demystifying trial networks and network meta-analysis. *BMJ* 2013;**346**:f2914. doi:10.1136/bmj.f2914

12 Dias S, Welton NJ, Sutton AJ, *et al. NICE DSU Technical Support Document 2: A Generalised Linear Modelling Framework for Pairwise and Network Meta-Analysis of Randomised Controlled Trials*. London: : National Institute for Health and Care Excellence (NICE) 2014. http://www.ncbi.nlm.nih.gov/books/NBK310366/ (accessed 8 Dec 2020).

13 Dias S, editor. *Network meta-analysis for decision making*. Hoboken, NJ: : Wiley 2018.

14 Tagliaferri SD, Miller CT, Ford JJ, *et al.* Randomized Trial of General Strength and Conditioning Versus Motor Control and Manual Therapy for Chronic Low Back Pain on Physical and Self-Report Outcomes. *J Clin Med* 2020;**9**. doi:10.3390/jcm9061726

15 Owen PJ, Miller CT, Mundell NL, *et al.* Which specific modes of exercise training are most effective for treating low back pain? Network meta-analysis. *Br J Sports Med* 2019;:in press. doi:10.1136/bjsports-2019-100886

16 Huang R, Ning J, Chuter VH, *et al.* Exercise alone and exercise combined with education both prevent episodes of low back pain and related absenteeism: systematic review and network meta-analysis of randomised controlled trials (RCTs) aimed at preventing back pain. *Br J Sports Med* Published Online First: 31 October 2019. doi:10.1136/bjsports-2018-100035

17 Arts MP, Kuršumović A, Miller LE, *et al.* Comparison of treatments for lumbar disc herniation: Systematic review with network meta-analysis. *Medicine (Baltimore)* 2019;**98**:e14410. doi:10.1097/MD.0000000000014410

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18 Guo J-R, Jin X-J, Shen H-C, *et al.* A Comparison of the Efficacy and Tolerability of the Treatments for Sciatica: A Network Meta-Analysis. *Ann Pharmacother* 2017;**51**:1041–52. doi:10.1177/1060028017722008

19 Bardin LD, King P, Maher CG. Diagnostic triage for low back pain: a practical approach for primary care. *Med J Aust* 2017;**206**:268–73. doi:10.5694/mja16.00828

20 Moher D, Liberati A, Tetzlaff J, *et al.* Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;**6**:e1000097. doi:10.1371/journal.pmed.1000097

21 Hutton B, Salanti G, Caldwell DM, *et al.* The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Ann Intern Med* 2015;**162**:777–84. doi:10.7326/M14-2385

22 Moher D, Shamseer L, Clarke M, *et al.* Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. *2015*;**4**:1. doi:10/gcsk56

23 Nussbaumer-Streit B, Klerings I, Dobrescu AI, *et al.* Excluding non-English publications from evidence-syntheses did not change conclusions: a meta-epidemiological study. *J Clin Epidemiol* 2020;**118**:42–54. doi:10/ghpwpp

24 Higgins J, Thomas J, Chandler J, *et al.*, editors. *Cochrane handbook for systematic reviews of interventions version 6.2 (updated February 2021)*. Cochrane 2021. https://training.cochrane.org/handbook (accessed 16 Aug 2021).

25 Morrison A, Polisena J, Husereau D, *et al.* The effect of English-language restriction on systematic review-based meta-analyses: a systematic review of empirical studies. *Int J Technol Assess Health Care* 2012;**28**:138–44. doi:10.1017/S0266462312000086

26 Koes BW, van Tulder M, Lin C-WC, *et al.* An updated overview of clinical guidelines for the management of non-specific low back pain in primary care. *Eur Spine J* 2010;**19**:2075–94. doi:10.1007/s00586-010-1502-y

27 van Tulder M, Becker A, Bekkering T, *et al.* Chapter 3. European guidelines for the management of acute nonspecific low back pain in primary care. *Eur Spine J* 2006;**15 Suppl 2**:S169-191. doi:10.1007/s00586-006-1071-2

28 Clancy C, Quinn A, Wilson F. The aetiologies of Failed Back Surgery Syndrome: A systematic review. *J Back Musculoskelet Rehabil* 2017;**30**:395–402. doi:10.3233/BMR-150318

29 Qaseem A, Wilt TJ, McLean RM, *et al.* Noninvasive Treatments for Acute, Subacute, and Chronic Low Back Pain: A Clinical Practice Guideline From the American College of Physicians. *Ann Intern Med* 2017;**166**:514–30. doi:10.7326/M16-2367

30 Cochrane Back and Neck. Our reviews. 2020.https://back.cochrane.org/our-reviews (accessed 17 Jun 2020).

31 Furlan AD, Malmivaara A, Chou R, *et al.* 2015 Updated Method Guideline for Systematic Reviews in the Cochrane Back and Neck Group. *Spine* 2015;**40**:1660–73. doi:10.1097/BRS.0000000000001061

32 Wan X, Wang W, Liu J, *et al.* Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol* 2014;**14**:135. doi:10/f7xmgr

33 Vucic K, Jelicic Kadic A, Puljak L. Survey of Cochrane protocols found methods for data extraction from figures not mentioned or unclear. *J Clin Epidemiol* 2015;**68**:1161–4. doi:10.1016/j.jclinepi.2014.11.016

34 Higgins JPT, Altman DG, Gotzsche PC, *et al.* The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;**343**:d5928–d5928. doi:10/fcq7fv

35 Higgins J, Deeks JJ, Altman DG. Chapter 16: Assessing risk of bias in cluster-randomized trials. In: Higgins J, Green, S, eds. *Cochrane Handbook for Systematic Reviews*

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of Interventions. The Cochrane Collaboration 2011. https://handbook-5-1.cochrane.org/chapter_16/16_3_2_assessing_risk_of_bias_in_cluster_randomized_trials.htm (accessed 16 Jul 2021).

36 Sterne JAC, Savović J, Page MJ, *et al.* RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;:l4898. doi:10/ggmfc5

37 Norman GR, Sloan JA, Wyrwich KW. Interpretation of changes in health-related quality of life: the remarkable universality of half a standard deviation. *Med Care* 2003;**41**:582–92. doi:10/dhstrr

38 Nikolakopoulou A, Higgins JPT, Papakonstantinou T, *et al.* CINeMA: An approach for assessing confidence in the results of a network meta-analysis. *PLOS Med* 2020;**17**:e1003082. doi:10/ggthw8

39 Schünemann H, Higgins J, GE Vist, *et al.* Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins J, Thomas J, Chandler J, *et al.*, eds. *Cochrane Handbook for Systematic Reviews of Interventions*. The Cochrane Collaboration 2021. https://training.cochrane.org/handbook/current/chapter-14 (accessed 16 Aug 2021).

40 Savović J, Turner RM, Mawdsley D, *et al.* Association Between Risk-of-Bias Assessments and Results of Randomized Trials in Cochrane Reviews: The ROBES Meta-Epidemiologic Study. *Am J Epidemiol* 2018;**187**:1113–22. doi:10/gdgm64

41 Armijo-Olivo S, Fuentes J, da Costa BR, *et al.* Blinding in Physical Therapy Trials and Its Association with Treatment Effects: A Meta-epidemiological Study. *Am J Phys Med Rehabil* 2017;**96**:34–44. doi:10/gmhbgv

42 Armijo-Olivo S, Saltaji H, Costa BR da, *et al.* What is the influence of randomisation sequence generation and allocation concealment on treatment effects of physical therapy trials? A meta-epidemiological study. *BMJ Open* 2015;**5**:e008562. doi:10/gb5db8

43 Puhan MA, Schünemann HJ, Murad MH, *et al.* A GRADE Working Group approach for rating the quality of treatment effect estimates from network meta-analysis. *BMJ* 2014;**349**:g5630. doi:10/gf9bh2

44 Brignardello-Petersen R, Bonner A, Alexander PE, *et al.* Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis. *J Clin Epidemiol* 2018;**93**:36–44. doi:10/gcz4pg

45 Sterne JA, Gavaghan D, Egger M. Publication and related bias in meta-analysis: power of statistical tests and prevalence in the literature. *J Clin Epidemiol* 2000;**53**:1119–29. doi:10/fjt66s

46 Daly C, Welton NJ, Dias S, *et al.* Meta-Analysis of Continuous Outcomes. NICE Guidelines Technical Support Unit http://www.bristol.ac.uk/media-library/sites/social-community-medicine/documents/mpes/gmd-2-continuous-jan2021.pdf

47 Marinho VC, Higgins JP, Sheiham A, *et al.* Fluoride toothpastes for preventing dental caries in children and adolescents. *Cochrane Database Syst Rev* 2003;:CD002278. doi:10/c9rtvd

48 Higgins JP, Eldridge S, Li T. Chapter 23: Including variants on randomized trials. In: Higgins J, Thomas J, Chandler J, *et al.*, eds. *Cochrane Handbook for Systematic Reviews of Interventions*. Cochrane 2021. https://training.cochrane.org/handbook/current/chapter-23 (accessed 16 Jul 2021).

49 Phillippo D. *dmphillippo/multinma: v0.3.0*. Zenodo 2021. doi:10.5281/zenodo.4617744

50 Pedder H. *MBNMAtime: Run time-course MBNMA models*. https://CRAN.R-project.org/package=MBNMAtime

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51 Pedder H, Dias S, Bennetts M, *et al.* Modelling time-course relationships with multiple treatments: Model-based network meta-analysis for continuous summary outcomes. *Res Synth Methods* 2019;**10**:267–86. doi:10/ghnwjc

52 Spiegelhalter DJ, Best NG, Carlin BP, *et al.* Bayesian measures of model complexity and fit. *J R Stat Soc Ser B Stat Methodol* 2002;**64**:583–639. doi:10/dfrgt6

53 Viechtbauer W. Conducting Meta-Analyses in R with the metafor Package. *J Stat Softw* 2010;**36**. doi:10/gckfpj

54 Viechtbauer W, Cheung MW-L. Outlier and influence diagnostics for meta-analysis. *Res Synth Methods* 2010;**1**:112–25. doi:10/fs53xp

55 Boonstra AM, Reneman MF, Waaksma BR, *et al.* Predictors of multidisciplinary treatment outcome in patients with chronic musculoskeletal pain. *Disabil Rehabil* 2015;**37**:1242–50. doi:10.3109/09638288.2014.961657

56 Cecchi F, Pasquini G, Paperini A, *et al.* Predictors of response to exercise therapy for chronic low back pain: result of a prospective study with one year follow-up. *Eur J Phys Rehabil Med* 2014;**50**:143–51.

57 van der Hulst M, Vollenbroek-Hutten MMR, Ijzerman MJ. A systematic review of sociodemographic, physical, and psychological predictors of multidisciplinary rehabilitation-or, back school treatment outcome in patients with chronic low back pain. *Spine* 2005;**30**:813–25. doi:10.1097/01.brs.0000157414.47713.78

58 Donegan S, Williamson P, D'Alessandro U, *et al.* Assessing key assumptions of network meta-analysis: a review of methods: Assessing assumptions of network meta-analysis. *Res Synth Methods* 2013;**4**:291–323. doi:10/gf8ddc

59 Dias S, Welton NJ, Sutton AJ, *et al. NICE DSU Technical Support Document 4: Inconsistency in Networks of Evidence Based on Randomised Controlled Trials*. London: : National Institute for Health and Care Excellence (NICE) 2014. http://www.ncbi.nlm.nih.gov/books/NBK310372/ (accessed 8 Dec 2020).

60 van Valkenhoef G, Dias S, Ades AE, *et al.* Automated generation of node-splitting models for assessment of inconsistency in network meta-analysis. *Res Synth Methods* 2016;**7**:80–93. doi:10/f8t6mt

61 Wade R, Sharif-Hurst S, Dias S. Patient characteristics as effect modifiers for psoriasis biologic treatment response: an assessment using network meta-analysis subgroups. *Syst Rev* 2020;**9**:132. doi:10/gmc79b

62 Faltinsen EG, Storebø OJ, Jakobsen JC, *et al.* Network meta-analysis: the highest level of medical evidence? *BMJ Evid-Based Med* 2018;**23**:56–9. doi:10/ghn2kn

63 Bagg MK, McLachlan AJ, Maher CG, *et al.* Paracetamol, NSAIDS and opioid analgesics for chronic low back pain: a network meta-analysis. *Cochrane Database Syst Rev* 2018;**2018**:CD013045. doi:10/gjkmg7

64 Migliorini F, Maffulli N, Eschweiler J, *et al.* Non-steroidal anti-inflammatory drugs and gabapentinoids for chronic lumbar pain: a Bayesian network meta-analysis of randomized controlled trials. *Br Med Bull* 2021;**138**:85–95. doi:10.1093/bmb/ldab003

65 Migliorini F, Maffulli N, Baroncini A, *et al.* Opioids for chronic low back pain management: a Bayesian network meta-analysis. *Expert Rev Clin Pharmacol* 2021;**14**:635–41. doi:10.1080/17512433.2021.1903316

66 Boya C, Bansal D, Kanakagiri S, *et al.* Efficacy and Safety of Opioid Analgesics for the Management of Chronic Low Back Pain: An Evidence from Bayesian Network Meta-Analysis. *Pain Physician* 2021;**24**:73–82.

67 Anheyer D, Klose P, Koch AK, *et al.* Comparative efficacy of different exercise interventions in chronic non-specific low back pain: protocol of a systematic review and network meta-analysis. *BMJ Open* 2020;**10**:e036050. doi:10.1136/bmjopen-2019-036050

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

68 Gao C, Chen G, Yang H, *et al.* Relative effectiveness of different forms of exercises for treatment of chronic low back pain: protocol for a systematic review incorporating Bayesian network meta-analysis. *BMJ Open* 2019;**9**:e025971. doi:10.1136/bmjopen-2018-025971

69 Yu H, Wang H, Ma T, *et al.* TCM nonpharmacological interventions for chronic low-back pain: A protocol for systematic review and network meta-analysis. *Medicine (Baltimore)* 2020;**99**:e22547. doi:10/ghmvnf

70 Ikemoto T, Miki K, Matsubara T, *et al.* Psychological Treatment Strategy for Chronic Low Back Pain. *Spine Surg Relat Res* 2018;**3**:199–206. doi:10/gmhbgw

71 Sun W, Zhang H, Lv C, *et al.* Comparative efficacy of 12 non-drug interventions on non-specific chronic low back pain in nurses: A systematic review and network meta-analysis. *J Back Musculoskelet Rehabil* 2021;**34**:499–510. doi:10/gmc79c

72 Rihn JA, Radcliff K, Norvell DC, *et al.* Comparative Effectiveness of Treatments for Chronic Low Back Pain: A Multiple Treatment Comparison Analysis. *Clin Spine Surg* 2017;**30**:204–25. doi:10/gjkk7d

73 Maher C, Underwood M, Buchbinder R. Non-specific low back pain. *Lancet Lond Engl* 2017;**389**:736–47. doi:10.1016/S0140-6736(16)30970-9

74 Oliveira CB, Maher CG, Pinto RZ, *et al.* Clinical practice guidelines for the management of non-specific low back pain in primary care: an updated overview. *Eur Spine J* 2018;**27**:2791–803. doi:10/gfp76w

75 Tagliaferri SD, Angelova M, Zhao X, *et al.* Artificial intelligence to improve back pain outcomes and lessons learnt from clinical classification approaches: three systematic reviews. *Npj Digit Med* 2020;**3**:1–16. doi:10/ghn2mv

76 Thompson T, Dias S, Poulter D, *et al.* Efficacy and acceptability of pharmacological and non-pharmacological interventions for non-specific chronic low back pain: a protocol for a systematic review and network meta-analysis. *Syst Rev* 2020;**9**:130. doi:10/ghmvnm

77 Low back pain and sciatica in over 16s: assessment and management. https://www.nice.org.uk/guidance/ng59 (accessed 31 Jul 2021).

78 Welton NJ, Caldwell DM, Adamopoulos E, *et al.* Mixed treatment comparison meta-analysis of complex interventions: psychological interventions in coronary heart disease. *Am J Epidemiol* 2009;**169**:1158–65. doi:10/c29ps8

79 Rücker G, Petropoulou M, Schwarzer G. Network meta-analysis of multicomponent interventions. *Biom J Biom Z* 2020;**62**:808–21. doi:10/ggd9ff

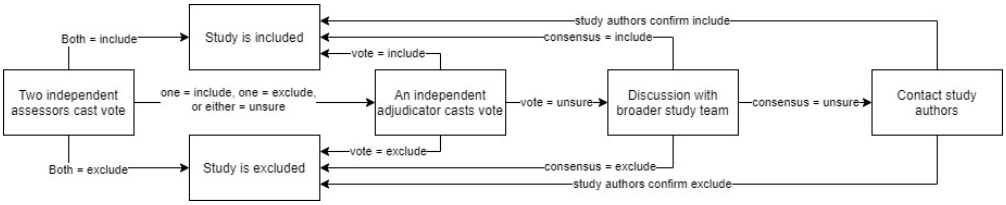
80 Stevens JW, Fletcher C, Downey G, *et al.* A review of methods for comparing treatments evaluated in studies that form disconnected networks of evidence. *Res Synth Methods* 2018;**9**:148–62. doi:10/gcpj92

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Figure 1. The process for determining study inclusion/exclusion.

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Process for determining study inclusion/exclusion.

324x63mm (72 x 72 DPI)

Supplemental Data A: Definitions of interventions and primary/secondary interventions

Acupuncture (acu)

Per prior Cochrane review [1], the definition of acupuncture used was "*the diagnosis was made using traditional acupuncture theory and the needles were inserted in classical meridian points, extra points or ah-shi points (painful points)*". Dry needling was classified with acupuncture and required needles to be inserted into myofascial trigger points. Acupressure, laser acupuncture and acupuncture via electrical stimulation were excluded from this acupuncture group as they did not involve needling. These interventions were included, respectively, under massage (acupressure) and electrotherapies (laser and electrical acupuncture).

Treatments within Class:

- acu_need: acupuncture following (traditional) acupuncture theory
- acu_dry: dry needling

Education (edu)

Patient education has been defined [2] previously "*a systematic experience, in a one-to one situation, that consists of one or more methods, such as the provision of information and advice and behaviour modification techniques*". Similar to this prior review, we considered education to occur when back pain patients were given information to help them understand their condition, what behaviours are likely to be more beneficial. 'Back school' interventions were considered education. Advice to stay active was considered education. Both group and individual education were included. Using brochure or booklet with education material was included if a clinician explained the information to the patient. Studies on instructions as to how to perform other kinds of interventions (e.g., how to do exercise, or were included, studies on instructions on how to perform exercises were not included.

Treatments within Class:

- edu_school: back school
- edu_pne: pain neuroscience education
- edu_book: via printed materials
- edu_grpind: remaining group and individual education

Electrophysical agents (elc)

Therapeutic heat and cold, laser (including laser acupuncture) and light therapies, classic electrotherapies (e.g., electrical stimulation modalities including TENS; electrical acupuncture also included here), various electromagnetic applications (e.g., pulsed shortwave therapy), ultrasound therapy and a variety of mechanical therapies (e.g., vibration therapy and intermittent pneumatic compression therapy) are included as electrophysical agents given these modalities are considered comparable [3]. The electrophysical agents must be applied externally without breaking or piercing the skin.

Whole body vibration, where a person experiences vibration through their whole body, is excluded.

Treatments within Class:

- elc_electric: electrical stim or input of some form
- elc_hot: heat
- elc_cold: cold

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- elc_mech: ultrasound therapy and a variety of mechanical therapies
- elc_etc: magnetic and remaining included

Epidural injections (epi)

As per prior Cochrane review [4], epidural injections involve the delivery of corticosteroid medication to the epidural space via injection. The anatomical approaches considered included, but were not limited to: caudal, interlaminar, and transforaminal approaches.

Treatments within Class:

- epi_caud: caudal approach
- epi_inter: interlaminar approach
- epi_trans: transforaminal approach
- epi_other: other included epidural INT not included in anatomical approaches listed above

Exercise (exe)

Exercise therapy has been [5] defined as "*a series of specific movements with the aim of training or developing the body by a routine practice or as physical training to promote good physical health*". We required that a clinician or study investigator instructed and/or prescribed exercises to patients with the goal of improving the patient's back disorder. Exercise could be performed as a group or individually. Whole body vibration and whole body vibration exercise was excluded.

Treatments within Class:

- exe_res: resistance exericse
- exe_sta: stabilization_motor_control
- exe_eso: pilates, yoga, traditional eastern approaches
- exe_aer: aerobic (e.g cycling, walking)
- exe_str: streching
- exe_oth: other and water based

Manual therapies and manipulation (man)

A prior Cochrane review [6] defined mobilisation as the "*use low-grade velocity, small or large amplitude passive movement techniques within the patient's range of motion and control*" and manipulation as "*a high velocity impulse or thrust applied to a synovial joint over a short amplitude at or near the end of the passive or physiologic range of motion, which is often accompanied by an audible crack*". The term "adjustments" is sometimes used in conjunction with chiropractic or osteopathic manual therapy treatment. Studies that incorporate visceral techniques as part of an osteopathy intervention will be included.

Treatments within Class:

- man_man: manual therapy and mobilisation (without manipulation)
- man_mip: manipulation
- man_chos: chiropractic or osteopathy not otherwise more precisely specified

Massage (mas)

Massage has been [7] defined as "*the manipulation of the soft tissue of whole body areas to bring about generalised improvements in health, such as relaxation or improved sleep, or*

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specific physical benefits, such as relief of muscular aches and pains" Trigger point therapy, myofascial release, Shiatsu, reflexology, and acupressure are also classified as massage.

Treatments within Class:

- mas_mas: massage
- mas_tpm: Trigger point therapy, myofascial release
- mas_oth: Shiatsu, reflexology, acupressure and other specifically named treatments determined to be massage

McKenzie (mck)

The McKenzie method [8] has also been termed Mechanical Diagnosis and Therapy and is a system that involves the use of mechanical loading strategies to guide specific treatment based on the patient's responses to these mechanical loading strategies (sub-group membership) [9]. In this treatment approach, treatment is individualized for each patient based on the response of their pain/impairment to mechanical loading strategies (sustained or repeated movements and postures) and classified into dysfunction, posture and derangement syndromes. Given it is the most prevalent classification, studies using directional preference treatment only (for derangement syndrome) will also be included. Directional preference management was defined as individualized treatment based on the response to mechanical loading strategies. Trials evaluating the effect of directional preference management on back pain were included.

Treatments within Class:

- mck_mck: Mckenzie

Pharmacotherapy (pha)

Pharmacotherapy interventions considered in this review included non-steroidal anti-inflammatory drugs (NSAIDs), Opioids, Skeletal muscle relaxants, Benzodiazepines, Antidepressants, Acetaminophen (paracetamol), systemic corticosteroids and anticonvulsants. Analgesic medicines work in various ways to reduce the intensity of pain but may also cause unwanted harmful effects.

Treatments within Class:

- pha_nsai: NSAIDs
- pha_opi: Opioids
- pha_relx: Skeletal muscle relaxants
- pha_benz:Benzodiazepines
- pha_antd: Antidepressants
- pha_para: Acetaminophen (paracetamol)
- pha_cort: systemic corticosteroids and
- pha_conv: anticonvulsants

Psychological therapies (including cognitive-behavioural therapies) (psy)

Per prior Cochrane review [10], psychological interventions were classed as any intervention that is designed following a psychological theory of behaviour and behaviour change. Mindfulness meditation, or other forms of meditation, were not, by themselves, considered psychological therapies.

Treatments within Class:

- psy_cbt: cognitive behavioural therapies

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- psy_oth: other psychological therapies

Percutaneous procedures (per)

The following percutaneous procedures were considered:

- Radio frequency denervation: Radiofrequency denervation has been defined [11] as "*a minimally invasive and percutaneous procedure performed under local anaesthesia or light intravenous sedation. Radiofrequency energy is delivered along an insulated needle in contact with the target nerves to denature the nerve*". It was initially developed for the lumbar zygopophyseal joint, and is now applied to denervate other joints in the spine [12].
- Spinal cord stimulation: "*This method was a clinical outgrowth from the well-known gate-control theory for segmental pain suppression. The idea was to apply electric stimulation to the dorsal columns of the spinal cord which are easily accessible and contain large diameter afferent fibers. Thus, stimulating electrodes were applied epidurally to the dorsal aspect of the cord. The gate control theory implied that activation of these coarse fiber systems inhibited transmission of nociceptive information at the segmental level and actually predicted that all types of pain would be equally suppressed*." [13,14]
- Percutaneous multifidus stimulation Percutaneous multifidus stimulation involves "*a stimulating probe is placed into the multifidus muscle via percutaneous procedure, using known anatomical landmarks to target the medial branches of the dorsal rami. Electrical stimulation is applied to target the medial branch of the dorsal ramus after the branch exits the intervertebral foramen prior to innervation of the multifidus and facet joints*."[15]
- Percutaneous rhizolysis, medial bundle branch blocks:
    - Percutaneous rhizolysis (radiofrequency neurotomy), medial bundle branch blocks: "Low-back pain may arise from degenerative changes in the posterior joints of the lumbar spine. These joints are innervated by a branch of the posterior primary ramus, which follows an anatomically constant course. Pain impulses from these joints can be interrupted by coagulating the nerve with a radiofrequency wave, the probe having been placed in the area of the nerve percutaneously." [16]
    - Facet joint medial bundle branch radiofrequency ablation (MBB-RFA) "*involves using energy in the radiofrequency range to perform necrosis of specific nerves (medial branches of the dorsal rami in patients with lumbar facetogenic pain), avoiding the neural transmission of pain. The aim of MBB-RFA is to both provide relief of pain and decrease the possibility of recurrence*". [17,18]

Treatments within Class:

- per_rad: Radio frequency denervation
- per_ssc: Spinal cord stimulation
- per_mfs: Percutaneous multifidus stimulation
- per_rhi: Percutaneous rhizolysis (radiofrequency neurotomy), medial bundle branch blocks
- per_mmb: Facet joint medial bundle branch radiofrequency ablation (MBB-RFA)

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Surgery (sur)

The following types of surgery were included:

- Discectomy (any type): open discectomy, sequestrectomy or aggressive discectomy, microdiscectomy, endoscopic open/percutaneous discectomy, automated open/percutaneous discectomy
- Non fusion stabilization [19]: Graf ligament, Dynesys, interspinous stabilisation devices (e.g., Coflex, Wallis ligament, DIAM), total disc arthroplasty (replacement), facet arthroplasty/facet replacement
- Fusion [19]: anterior, posterior, or circumferential spinal fusion (decompression/discectomy/laminectomy/laminotomy) with/without autologous bone graft harvested from the iliac crest or use of allograft femoral rings stuffed with autologous cancellous bone with/without pedicle screw [20]

Surgery may include indirect/direct decompression [21], decompression with/without instrumentation fusion [21,22] PLIF, ALIF, TLIF, minimally invasive spine surgeries (including laparoscopic ALIF, minimally invasive PLIF, XLIF, OLIF, AxiaLIF).

Treatments within Class:

- sur_dis: Discectomy (any type)
- sur_nstab: Non fusion stabilization
- sur_fus: Fusion
- sur_deco: Decompression/laminectomy/laminotomy without an instrument for foraminal/canal stenosis

Traction (tra)

Traction involves application of a distractive axial force to the spine and trunk for therapeutic effect [23]: "*Mechanical or motorized traction (where the traction is exerted by a motorized pulley), manual traction (in which the traction is exerted by the therapist, using his or her body weight to alter the force and direction of the pull), and auto-traction (where the person controls the traction forces by grasping and pulling bars at the head of the traction table)*" [23] were included as traction. Other forms of traction may include the use of gravity to generate the traction force (e.g., on a tilted table, or hung vertically by the lower extremities).

Treatments within Class:

- tra_mech: Mechanical or motorized traction
- tra_man: manual traction
- tra_auto: auto-traction and use of gravity to generate the traction force

Multidisciplinary (multidisciplinary pain management) (mul)

Multidisciplinary pain management incorporates a number of intervention types, such as education (e.g., mechanisms of chronic pain, anatomy), goal setting, exercise, stress management, relaxation and imagery, meditation and aspects of psychological therapies, medication management, family member participation implemented as one package of treatment [24–26]. These may be done as individual sessions or as group sessions. If a study labelled its intervention as multidisciplinary pain management, then this was considered multidisciplinary pain management. Other studies may have combined individual interventions

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into a multidisciplinary program but did not specifically label it as multidisciplinary pain management. In this case, if the reviewers agreed that the intervention included a minimum of education, exercise, psychological therapies delivered by a multidisciplinary clinician team (at least 2 clinicians from different fields), this was classified as 'multidisciplinary pain management'. Otherwise, these interventions were classified under 'treatment combinations' (below).

Treatments within Class:

- mul_mdp: Multidisciplinary pain management

Physical therapy (otherwise not falling into specific treatment combination) (pio)
Into this group fall any interventions that are generic 'physiotherapy' or 'physical therapy' treatments, often at the discretion of the clinician, but otherwise not detailed or defined.

Treatments within Class:

- pio_pio: generic physiotherapy or physical therapy treatments

Placebo or sham (pla)
Any intervention defined as a placebo or sham intervention by the study authors, or described as such consistent with previous meta-analysis [27].

Treatments within Class:

- pla_pla: placebo

"Usual care" (e.g., GP Management) (usu)
Intervention deemed 'usual care', including GP management.

Treatments within Class:

- usu_usu: usual care

No treatment (true control) (tru)
No intervention provided, including waitlist control where no treatment is given.

Treatments within Class:

- tru_tru: true control, no intervention
- tru_wait: waiting list control where not treatment is given

Combinations of the above treatments were included and classified according to their primary and secondary treatment components via agreement between the extractors (with adjudication where necessary)

Definition of primary and secondary INT components
The following approach was used to classify primary and secondary intervention components in groups that receive multiple treatments within the same treatment group but did not clearly fall under the multidisciplinary definition above:

1) Pick the primary intervention that contributes to the treatment group: if an intervention comprised >50% of the treatment (per judgement of the extractor), then it was taken as 'primary'. If no intervention component was >50%, then pick the one with the highest proportion.

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In cases that were unclear, the following hierarchy of guiding principles was used:

- Any prior publications (e.g., protocol paper, primary outcome publication) arising from the same study were checked.
- A treatment component that is more thoroughly described could be considered the primary component. For example, if exercise was fully described but advice is labelled as "advice" and not described in similar detail, then exercise was considered as the primary.
- Where a treatment component was mentioned in either the article title or the group subheading was labelled as one of the interventions, then that was taken as the primary intervention component. For example, if the group subheading was called "exercise" but it contained exercise and advice components, then exercise was considered the primary component).
- To split true stalemates, the intervention element mentioned first in the treatment description and/or label was taken to be the primary component (e.g., "exercise and advice" = exercise mentioned first and therefore primary component).
- A minimum threshold to be classified as a primary component was 25%.

2) Secondary component of treatments with multiple components: in some cases, a treatment group may have more than two components, but not fall under the multidisciplinary definition. In this case, the following principles were followed: only ONE secondary intervention component was included, regardless of how many there were. To qualify as being classified as a secondary treatment component, it needed to represent at least 20% of the total intervention (per judgement of the extractor), otherwise the intervention will be classified as having only a primary intervention with no secondary component.

3) If a study arm could not be classified according to the above criteria, then it was treated as a non-included INT (see below). We considered including an additional 'multimodal' category beyond the multidisciplinary group defined above. However, we determined this would be uninformative as it would encompass a heterogeneous range of treatments and thus not provide useful guidance for clinical practice.

Where both primary and secondary intervention components are present we will include these in analyses as combinations of intervention and they will be analysed separately. For example, Physical therapy as a primary component and Massage as a secondary component will be analysed as “Physical therapy + massage”. Due to the approach we have described for classifying primary and secondary components, the order of components may be important, such that we assume that “Physical therapy + massage” is not the same as “Massage + physical therapy”.

Where a secondary intervention component is given in all arms of a study, in addition to the analysis above we will also fit a model in which the study treatments are coded as only the primary intervention in order to test whether assuming additivity of treatment efficacy is reasonable, as this may lead to better connected NMAs with more precise estimates.

Handling of studies that examined non-included INTs

Some studies will examine an INT that is not subject of the current review. In this case, the arms in the study were assessed on a case by case basis.

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- If the 'primary treatment component' of an arm was a non-included INT (e.g., back belts), then that individual arm was not included in extraction and therefore analysis.
- If the 'primary treatment component' was an included INT but the 'secondary treatment component' a non-included INT, then the individual arm was included.
- Pending these decisions, if at least two arm of an individual study could be included, then the study as a whole was included. Otherwise it was excluded.

For example, in the case of a three arm study [28] on "back belt + exercise" vs "exercise" vs "control", the "back belt + exercise" arm was excluded, but the "exercise" and "control" arms were included. Thus the study could also be included.

1 Furlan AD, van Tulder MW, Cherkin D, *et al.* Acupuncture and dry-needling for low back pain. *Cochrane Database Syst Rev* Published Online First: 24 January 2005. doi:10.1002/14651858.CD001351.pub2

2 Engers AJ, Jellema P, Wensing M, *et al.* Individual patient education for low back pain. *Cochrane Database Syst Rev* Published Online First: 23 January 2008. doi:10.1002/14651858.CD004057.pub3

3 Watson T. Key concepts with electrophysical agents. *Phys Ther Rev* 2010;**15**:351–9. doi:10.1179/1743288X10Y.0000000009

4 Oliveira CB, Maher CG, Ferreira ML, *et al.* Epidural corticosteroid injections for lumbosacral radicular pain. *Cochrane Database Syst Rev* Published Online First: 9 April 2020. doi:10.1002/14651858.CD013577

5 Hayden JA, van Tulder MV, Malmivaara A, *et al.* Exercise therapy for treatment of non-specific low back pain. *Cochrane Database Syst Rev* 2005;:77. doi:10.1002/14651858.CD000335.pub2

6 Rubinstein SM, van Middelkoop M, Assendelft WJ, *et al.* Spinal manipulative therapy for chronic low-back pain. *Cochrane Database Syst Rev* Published Online First: 16 February 2011. doi:10.1002/14651858.CD008112.pub2

7 Furlan AD, Giraldo M, Baskwill A, *et al.* Massage for low-back pain. *Cochrane Database Syst Rev* Published Online First: 1 September 2015. doi:10.1002/14651858.CD001929.pub3

8 McKenzie R, May S. *The lumbar spine, volume one: mechanical diagnosis and therapy*. Wellington, N.Z: : Spinal Publications 2003.

9 Surkitt LD, Ford JJ, Hahne AJ, *et al.* Efficacy of Directional Preference Management for Low Back Pain: A Systematic Review. *Phys Ther* 2012;**92**:652–65. doi:10.2522/ptj.20100251

10 Williams AC de C, Fisher E, Hearn L, *et al.* Psychological therapies for the management of chronic pain (excluding headache) in adults. *Cochrane Database Syst Rev* Published Online First: 12 August 2020. doi:10.1002/14651858.CD007407.pub4

11 NICE Guidelines. Chapter 23 Radiofrequency denervation for facet joint pain. In: *Sciatica in Over 16s: Assessment and management*. London: : National Institute for Health and Care Excellence (UK) 2016.

12 Maas ET, Ostelo RW, Niemisto L, *et al.* Radiofrequency denervation for chronic low back pain. *Cochrane Database Syst Rev* Published Online First: 23 October 2015. doi:10.1002/14651858.CD008572.pub2

13 Linderoth B, Foreman RD. Physiology of Spinal Cord Stimulation: Review and Update. *Neuromodulation Technol Neural Interface* 1999;**2**:150–64. doi:10.1046/j.1525-1403.1999.00150.x

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14 Oakley JC, Prager JP. Spinal Cord Stimulation: Mechanisms of Action. *Spine* 2002;**27**:2574–83. doi:10.1097/00007632-200211150-00034

15 Cohen S, Gilmore C, Kapural L, *et al.* Percutaneous Peripheral Nerve Stimulation for Pain Reduction and Improvements in Functional Outcomes in Chronic Low Back Pain. *Mil Med* 2019;**184**:537–41. doi:10.1093/milmed/usy310

16 McCulloch JA. Percutaneous Radiofrequency Lumbar Rhizolysis (Rhizotomy). *Stereotact Funct Neurosurg* 1976;**39**:87–96. doi:10.1159/000102481

17 Shealy CN. Percutaneous radiofrequency denervation of spinal facets: Treatment for chronic back pain and sciatica. *J Neurosurg* 1975;**43**:448–51. doi:10.3171/jns.1975.43.4.0448

18 Akbary K, Kim J-S. Ablation of Medial Bundle Branch Under Spinal Endoscopy. In: Kim J-S, Lee JH, Ahn Y, eds. *Endoscopic Procedures on the Spine*. Singapore: : Springer Singapore 2020. 313–20. doi:10.1007/978-981-10-3905-8_24

19 Baliga S, Treon K, Craig NJA. Low Back Pain: Current Surgical Approaches. *Asian Spine J* 2015;**9**:645. doi:10.4184/asj.2015.9.4.645

20 Christensen FB, Bünger C. Stabilisation surgery for chronic low back pain: indications, surgical procedures, and outcome. *Scand J Rheumatol* 2004;**33**:210–7. doi:10.1080/03009740410005458

21 Watters WC, Bono CM, Gilbert TJ, *et al.* An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spondylolisthesis. *Spine J Off J North Am Spine Soc* 2009;**9**:609–14. doi:10.1016/j.spinee.2009.03.016

22 Kreiner DS, Shaffer WO, Baisden JL, *et al.* An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update). *Spine J Off J North Am Spine Soc* 2013;**13**:734–43. doi:10.1016/j.spinee.2012.11.059

23 Wegner I, Widyahening IS, van Tulder MW, *et al.* Traction for low-back pain with or without sciatica. *Cochrane Database Syst Rev* Published Online First: 19 August 2013. doi:10.1002/14651858.CD003010.pub5

24 Cassidy EL, Atherton RJ, Robertson N, *et al.* Mindfulness, functioning and catastrophizing after multidisciplinary pain management for chronic low back pain: *Pain* 2012;**153**:644–50. doi:10.1016/j.pain.2011.11.027

25 Maruta T, Swanson DW, McHardy MJ. Three year follow-up of patients with chronic pain who were treated in a multidisciplinary pain management center: *Pain* 1990;**41**:47–53. doi:10.1016/0304-3959(90)91108-U

26 Jensen MP, Turner JA, Romano JM. Changes after multidisciplinary pain treatment in patient pain beliefs and coping are associated with concurrent changes in patient functioning: *Pain* 2007;**131**:38–47. doi:10.1016/j.pain.2006.12.007

27 Hrobjartsson A, Gotzsche PC. Is the placebo powerless? Update of a systematic review with 52 new randomized trials comparing placebo with no treatment. *J Intern Med* 2004;**256**:91–100. doi:10.1111/j.1365-2796.2004.01355.x

28 Dalichau S, Scheele K. Auswirkungen elastischer Lumbal-Stützgurte auf den Effekt eines Muskeltrainingsprogrammes für Patienten mit chronischen Rückenschmerzen. *Z Für Orthop Ihre Grenzgeb* 2000;**138**:8–16. doi:10.1055/s-2000-10106

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Supplemental Data B: Search Strategy

Date of database search: 13.11.2019

MEDLINE

Search	Query	Hits
#1	back pain[MeSH Terms] OR low back pain[MeSH Terms] OR back pain*[Title/Abstract] OR lumb* pain[Title/Abstract] OR lumbago[Title/Abstract] OR backache*[Title/Abstract] OR back ache*[Title/Abstract] OR spinal stenosis[Title/Abstract] OR canal stenosis[Title/Abstract] OR lumbar stenosis[Title/Abstract] OR lateral stenosis[Title/Abstract] OR foramin stenosis[Title/Abstract] OR neurogenic claudication[Title/Abstract] OR radiculopathy[Title/Abstract] OR radicular pain[Title/Abstract] OR spondylolisthesis[Title/Abstract] OR spondylosis[Title/Abstract] OR sciatica[Title/Abstract] OR intervertebral disc displacement[Title/Abstract] OR referred pain[Title/Abstract] OR spinal nerve roots[Title/Abstract] OR neurologic signs[Title/Abstract] OR radiat* pain[Title/Abstract] OR radiat* symptoms[Title/Abstract] OR parathesia[Title/Abstract] OR numbness[Title/Abstract]	131336
#2	randomized[Title/Abstract] OR placebo[Title/Abstract] OR randomly[Title/Abstract] OR “drug therapy”[Title/Abstract] OR trial[Title/Abstract] OR groups[Title/Abstract]	2868072
#3	(animals[MeSH] NOT humans[MeSH]))	4659784
#4	#1 AND #2	25960
#5	#4 NOT #3	24928
#6	#5 AND Filters: Randomized Controlled Trial; Clinical Trial; Humans	7237

SPORTDiscus

Search	Query	Hits
#1	(DE “lumbar pain”) OR (DE backache) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	11187
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	197381
#4	#1 AND #2	2970

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CINAHL

Search	Query	Hits
#1	(MH low back pain) OR (MH back pain) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	45198
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	889333
#3	MH animals NOT MH human	74138
#4	#1 AND #2	11513
#5	#4 NOT #3	11461
#4	#5 AND Filters: Exclude MEDLINE records; Human; Randomized Controlled Trial	1335

PsycINFO

Search	Query	Hits
#1	(MA low back pain) OR (MA back pain) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness))	8813
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	1030813
#3	MA animals NOT MA human	196321
#4	#1 AND #2	2829

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#5	#4 NOT #3	2809
#6	#5 AND Filters: Human; Journal Article	2377

EMBASE

Search	Query	Hits
#1	'low back pain'/exp OR 'backache'/exp OR 'back pain*':ab,ti OR 'lumb* pain':ab,ti OR lumbago:ab,ti OR backache*:ab,ti OR 'back ache*':ab,ti OR 'spinal stenosis':ab,ti OR 'canal stenosis':ab,ti OR 'lumbar stenosis':ab,ti OR 'lateral stenosis':ab,ti OR 'foramin stenosis':ab,ti OR 'neurogenic claudication':ab,ti OR radiculopathy:ab,ti OR 'radicular pain':ab,ti OR spondylolisthesis:ab,ti OR spondylosis:ab,ti OR sciatica:ab,ti OR 'intervertebral disc displacement':ab,ti OR 'referred pain':ab,ti OR 'spinal nerve roots':ab,ti OR 'neurologic signs':ab,ti OR 'radiat* pain':ab,ti OR 'radiat* symptoms':ab,ti OR parathesia:ab,ti OR numbness:ab,ti	161402
#2	randomized:ab,ti OR placebo:ab,ti OR randomly:ab,ti OR "drug therapy":ab,ti OR trial:ab,ti OR groups:ab,ti	3957907
#3	'animal'/exp NOT 'human'/exp	5386039
#4	#1 AND #2	33606
#5	#4 NOT #3	32975
#6	#5 AND Filters: Controlled Clinical Trial; Randomized Controlled Trial; Exclude MEDLINE	2627

CENTRAL

Search	Query	Hits
#1	(MeSH descriptor: [back pain] explode all trees) OR (MeSH descriptor: [low back pain] explode all trees) OR ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR parathesia OR numbness):ti,ab,kw	3401
#2	(randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups):ti,ab,kw	1204707
#3	#1 AND #2	2895
#4	(MeSH descriptor: [Animals] explode all trees) NOT (MeSH descriptor: [Humans] explode all trees)	7286
#5	#4 NOT #3	2893
#6	#5 AND Filters: Exclude MEDLINE; Exclude EMBASE; Trials	456

TOTAL from data base searches (with duplicates): 17002

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Date of database search: 24/07/2020

MEDLINE

Search	Query	Hits
#1	back pain[MeSH Terms] OR low back pain[MeSH Terms] OR sciatica[MeSH Terms] OR back pain*[Title/Abstract] OR lumb* pain[Title/Abstract] OR lumbago[Title/Abstract] OR backache*[Title/Abstract] OR back ache*[Title/Abstract] OR spinal stenosis[Title/Abstract] OR canal stenosis[Title/Abstract] OR lumbar stenosis[Title/Abstract] OR lateral stenosis[Title/Abstract] OR foramin stenosis[Title/Abstract] OR neurogenic claudication[Title/Abstract] OR radiculopathy[Title/Abstract] OR radicular pain[Title/Abstract] OR spondylolisthesis[Title/Abstract] OR spondylosis[Title/Abstract] OR sciatica[Title/Abstract] OR intervertebral disc displacement[Title/Abstract] OR referred pain[Title/Abstract] OR spinal nerve roots[Title/Abstract] OR neurologic signs[Title/Abstract] OR radiat* pain[Title/Abstract] OR radiat* symptoms[Title/Abstract] OR paresthesia[Title/Abstract] OR paraesthesia[Title/Abstract] OR numbness[Title/Abstract]	141,803
#2	randomized[Title/Abstract] OR placebo[Title/Abstract] OR randomly[Title/Abstract] OR "drug therapy"[Title/Abstract] OR trial[Title/Abstract] OR groups[Title/Abstract]	2972235
#3	animals[MeSH] NOT humans[MeSH]	4720975
#4	#1 AND #2	28330
#5	#4 NOT #3	27250
#6	#5 AND Filters: Randomized Controlled Trial; Clinical Trial; Humans	9188

SPORTDiscus

Search	Query	Hits
#1	(DE "lumbar pain") OR (DE backache) OR (DE sciatica) OR (TI ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	14,427
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	237964
#4	#1 AND #2	4142

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CINAHL

Search	Query	Hits
#1	(MH low back pain) OR (MH back pain) OR (MH sciatica) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	52162
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	997530
#3	MH animals NOT MH human	79989
#4	#1 AND #2	13351
#5	#4 NOT #3	13289
#4	#5 AND Filters: Exclude MEDLINE records; Human; Randomized Controlled Trial	699

PsycINFO

Search	Query	Hits
#1	(MA low back pain) OR (MA back pain) OR (MA sciatica) OR (TI (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness) OR AB (“back pain*” OR “lumb* pain” OR lumbago OR backache* OR “back ache*” OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness))	9726
#2	TI (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups) OR AB (randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups)	1055873
#3	MA animals NOT MA human	197309
#4	#1 AND #2	3091

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#5	#4 NOT #3	3070
#6	#5 AND Filters: Human; Journal Article	2628

EMBASE

Search	Query	Hits
#1	'low back pain'/exp OR 'backache'/exp OR 'sciatica'/exp OR 'back pain*':ab,ti OR 'lumb* pain':ab,ti OR lumbago:ab,ti OR backache*:ab,ti OR 'back ache*':ab,ti OR 'spinal stenosis':ab,ti OR 'canal stenosis':ab,ti OR 'lumbar stenosis':ab,ti OR 'lateral stenosis':ab,ti OR 'foramin stenosis':ab,ti OR 'neurogenic claudication':ab,ti OR radiculopathy:ab,ti OR 'radicular pain':ab,ti OR spondylolisthesis:ab,ti OR spondylosis:ab,ti OR sciatica:ab,ti OR 'intervertebral disc displacement':ab,ti OR 'referred pain':ab,ti OR 'spinal nerve roots':ab,ti OR 'neurologic signs':ab,ti OR 'radiat* pain':ab,ti OR 'radiat* symptoms':ab,ti OR paresthesia:ab,ti OR paraesthesia:ab,ti OR numbness:ab,ti	176118
#2	randomized:ab,ti OR placebo:ab,ti OR randomly:ab,ti OR "drug therapy":ab,ti OR trial:ab,ti OR groups:ab,ti	4102141
#3	'animal'/exp NOT 'human'/exp	5464750
#4	#1 AND #2	37042
#5	#4 NOT #3	36356
#6	#5 AND Filters: Controlled Clinical Trial; Randomized Controlled Trial; Exclude MEDLINE	2718

CENTRAL

Search	Query	Hits
#1	(MeSH descriptor: [back pain] explode all trees) OR (MeSH descriptor: [low back pain] explode all trees) OR (MeSH descriptor: [sciatica] explode all trees) OR ("back pain*" OR "lumb* pain" OR lumbago OR backache* OR "back ache*" OR spinal stenosis OR canal stenosis OR lumbar stenosis OR lateral stenosis OR foramin stenosis OR neurogenic claudication OR radiculopathy OR radicular pain OR spondylolisthesis OR spondylosis OR sciatica OR intervertebral disc displacement OR referred pain OR spinal nerve roots OR neurologic signs OR radiat* pain OR radiat* symptoms OR paresthesia OR paraesthesia OR numbness):ti,ab,kw (Word variations have been searched):ti,ab,kw	23060
#2	(randomized OR placebo OR randomly OR "drug therapy" OR trial OR groups):ti,ab,kw	1228587
#3	(MeSH descriptor: [Animals] explode all trees) NOT (MeSH descriptor: [Humans] explode all trees)	606
#4	#1 AND #2	19342
#5	#4 NOT #3	19340
#6	#5 AND Filters: Exclude MEDLINE; Exclude EMBASE; Trials	1258

TOTAL from data base searches (with duplicates): 20633
TOTAL from prior systematic reviews (with duplicates): 1783

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TOTAL from reference lists of 17 relevant Cochrane reviews not included in reviews from last 10 years: 663
Duplicates removed (by Covidence): 3557
TOTAL for screening: 19522

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Supplemental Data C: GRADE Criteria

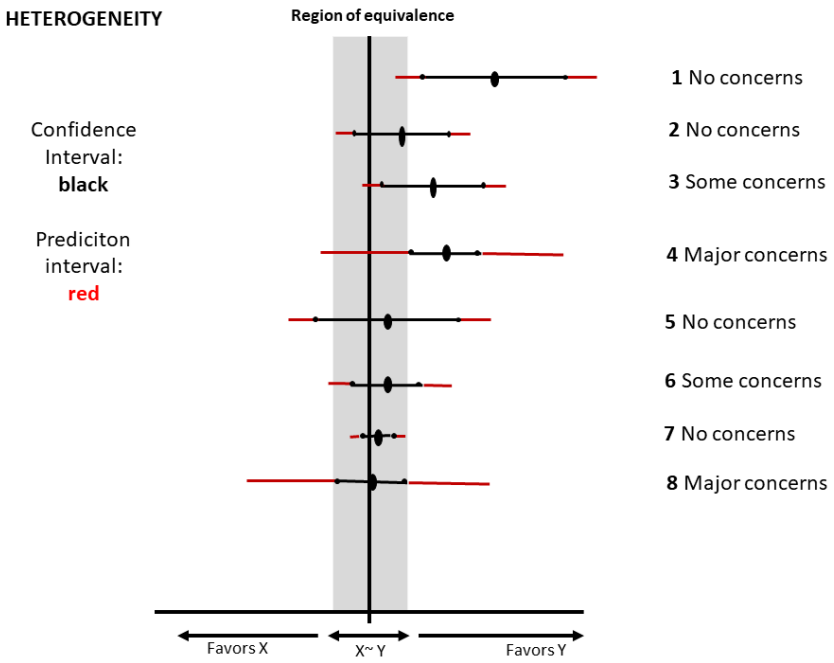
1. Limitations in study design – Cochrane Risk of Bias Tool 1.0

- Selection bias (random sequence generation, allocation concealment, group similarities at baseline);
- Performance bias (blinding of participants and/or healthcare providers);
- Attrition bias (drop outs and intention-to-treat analysis);
- Detection bias (blinding of the outcome assessors and timing of outcome assessment);
- Reporting bias (selective reporting).

We downgraded the quality of the evidence:

- By one level if >50% of participants were from studies with selection bias **and** performance bias.
- Inadequate randomization and lack of blinding may lead to an exaggeration of the intervention effect estimates [1–3].

Unexplained heterogeneity or inconsistency of results



- Pre-defined area/range of equivalence: We define a range of equivalence of SMD -0.5 to 0.5 [4].
- Downgrade two levels if there is a major concern and one level if there are some concerns.
- If there are very few trials, the amount of heterogeneity is poorly estimated and prediction intervals are unreliable, we will downgrade based on reference priors [5].

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Indirectness

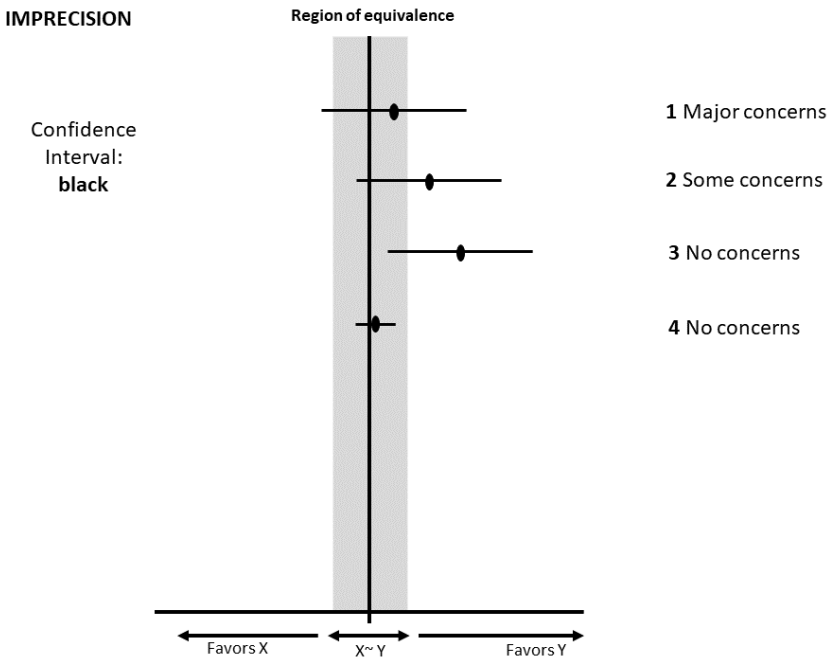
Domain (original question asked)	Description (evidence found and included, including evidence from other studies) – consider the domains of study design and study limitation, inconsistency, imprecision and publication bias	Judgment – is the evidence sufficiently direct?			
Population:		Yes	Probably yes	Probably no	No
Intervention:		Yes	Probably yes	Probably no	No
Comparator:		Yes	Probably yes	Probably no	No
Direct comparison:		Yes	Probably yes	Probably no	No
Outcome:		Yes	Probably yes	Probably no	No
Final judgement about indirectness across domains:	No indirectness ? => No downgrade. Serious indirectness ? => Downgrade one level. Very serious indirectness ? => Downgrade two levels.				

Two components for network meta-analysis:

- similarity of studies in the analysis to the target question (PICO)
- similarity of the studies in the analysis to each other (relates to transitivity assumption)

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Imprecision



- Downgrade two levels if there is a major concern and one level if there are some concerns.

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Publication bias [6]

Reporting bias may be suspected when the following occur:

- Prior documented evidence of reporting bias in trials in the field.
- meta-analysis is based on a small number of new studies, typically positive findings (e.g. new drugs may have positive findings early and later the true effect size becomes apparent).
- Industry-funded trials dominate
- Known unpublished data from grey literature not included.

Reporting bias is considered to not be present in the following situations:

- Analytical methods indicate the findings from small are similar to those in large/published studies
- Findings from unpublished studies agree with published studies.
- Prospective trial registration, protocol publication and/or clinical trial registries are used extensively in the field and do not indicate important discrepancies with published reports.

⇨ Downgrade one level if publication bias is suspected.

Criteria specific to NMA:

- Do **not consider imprecision** when rating the direct and indirect estimates to inform the rating of NMA estimates [7].
- **No need to rate the indirect evidence** when the certainty of the direct evidence is high and the contribution of the direct evidence to the network estimate is at least as great as that of the indirect evidence.

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1 Savović J, Turner RM, Mawdsley D, *et al.* Association Between Risk-of-Bias Assessments and Results of Randomized Trials in Cochrane Reviews: The ROBES Meta-Epidemiologic Study. *Am J Epidemiol* 2018;**187**:1113–22. doi:10/gdgm64

2 Armijo-Olivo S, Fuentes J, da Costa BR, *et al.* Blinding in Physical Therapy Trials and Its Association with Treatment Effects: A Meta-epidemiological Study. *Am J Phys Med Rehabil* 2017;**96**:34–44. doi:10/gmhbgv

3 Armijo-Olivo S, Saltaji H, Costa BR da, *et al.* What is the influence of randomisation sequence generation and allocation concealment on treatment effects of physical therapy trials? A meta-epidemiological study. *BMJ Open* 2015;**5**:e008562. doi:10/gb5db8

4 Norman GR, Sloan JA, Wyrwich KW. Interpretation of changes in health-related quality of life: the remarkable universality of half a standard deviation. *Med Care* 2003;**41**:582–92. doi:10/dhstrr

5 Rhodes KM, Turner RM, Higgins JPT. Predictive distributions were developed for the extent of heterogeneity in meta-analyses of continuous outcome data. *J Clin Epidemiol* 2015;**68**:52–60. doi:10/f6thkp

6 Nikolakopoulou A, Higgins JPT, Papakonstantinou T, *et al.* CINeMA: An approach for assessing confidence in the results of a network meta-analysis. *PLOS Med* 2020;**17**:e1003082. doi:10/ggthw8

7 Brignardello-Petersen R, Bonner A, Alexander PE, *et al.* Advances in the GRADE approach to rate the certainty in estimates from a network meta-analysis. *J Clin Epidemiol* 2018;**93**:36–44. doi:10/gcz4pg

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-P reporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a - not an update

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1-2
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	2
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a - not an amended version
Support			
Sources	#5a	Indicate sources of financial or other support for the review	2
Sponsor	#5b	Provide name for the review funder and / or sponsor	n/a - no sponsor
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a - no sponsor
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	5-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6–7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6; 9–10
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	9; (Supplemental data B, 36-42)
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through	9–10

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		each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	10–11
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	10
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	11–12
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	13-18
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	14–15

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Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	14-15
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	n/a
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	12

Notes:

- 1b: n/a - not an update
- 4: n/a - not an amended version
- 5b: n/a - no sponsor
- 5c: n/a - no sponsor
- 10: 27-33 (Supplemental data B) The PRISMA-P elaboration and explanation paper is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 17. March 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5–7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	9
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	9; Supplemental Data B, 36–42
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	9–10
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	10–11
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	10–11
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	10; 13
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	11–12
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	13
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	13-18
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	13; 17
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	13-18
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	13–15
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	17
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	13; 17
Reporting bias	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	12-13; 17-18

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	12-13
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	
Study characteristics	17	Cite each included study and present its characteristics.	
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	
	23b	Discuss any limitations of the evidence included in the review.	
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	
Competing interests	26	Declare any competing interests of review authors.	
Availability of data, code and	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
other materials			

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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